

DURGASOFT

Full Stack web developement by using Python and Django

Python + UI Technologies + Django

Fast track one month:

Videos

Study Materials

Videos ---> 6 months access

Course completion certificate

projects-->covered

2 to 4 years of exp

web application:

The application which will provide services over the web

gmail.com

facebook.com

1. Front end

2. Back end

1. Front end:

It represents what end user is seeing on the website.

We can develop front end components by using :

HTML,CSS,JS,BootStrap,jQuery etc

HTML,CSS,JS --->Basic Front end technologies

BootStrap,jQuery -->Advanced Front end libraries

HTML-->Hyper text markup language

It is mandatory to develop web applications.

It represents the structure of web page

CSS --->Cascading Style Sheets

To makeup our html components like adding colors,fonts,borders etc optional...

Java Script:

To add functionality to our html components

HTML --->meant for only static response

HTML+JS --->meant for dynamic response

Can we say, JavaScript is like dancing after drinking Beer(Action)?

static response: won't be changed from person to person and time to time.

Dynamic response: will be changed from person to person and time to time.

Bootstrap and jQuery

2. Back end:

It is technology which is used to decide what to show to end user on the front end.

List of trains from Hyderabad to Delhi

- 1.**
- 2.**
- 3.**

back end is responsible to generate required response

That response should be displayed to the end user by using front end.

back end: 3 components

1. Language: Java,Python,....
2. Framework: 70 to 90% of activities (Struts, JSF, Spring MVC etc), Django, Flask, Pyramid etc
3. Database: MongoDB, Oracle, MySQL etc

Biryani

We can prepare on our own(writing everything on your own without using any framework)

We can order online(using framework)

web application

front end--->What we are seeing

HTML, CSS, JS

back end-->generates what we have to see

Language-->Python

Framework: Django

Database-->

Rest api ????

9:15PM to 10:30 PM

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8096969696

Session-1: <https://www.youtube.com/watch?v=bNjNNtpfLQw>

Python(Core Python and Advanced Python)

Web development side:

UI Technologies

Django

Django Rest Framework

web application : gmail.com

facebook.com, amazon.in

1. Front-End

2. Back-End

1. Front-End: What user is seeing on the website

HTML,CSS,JS

Bootstrap,jQuery

2. Back end:

- 1. Language: Python**
- 2. Framework: Django, Flask, Pyramid etc 70% to 90%**
- 3. Database: SQLite, Oracle, MongoDB etc**

Java people

Python + Django

HTML:

gmail.com

HTML:

Hypertext markup language.

compulsory technology

amazon.in

spicejet.com

The main purpose of html is to define structure of the page.

Structure of HTML Page:

2 parts

1. Head

2. Body

1. Head:

It contains meta data like title,description,author,keywords etc
Here only we have to provide links to css files, js files

2. Body: Original data

PDF file--->Adobe Reader

HTML --->Browser(Chrome)

test.py

test.html

<HTML>

<head>

</head>

<body>

</body>

</html>

HTML Comments:

This is python comment

// This is java single line comment

<!-- This is html comment -->

Heading tags:

6 heading tags

<h1> This is h1 Heading </h1>

<h2> This is h2 Heading </h2>

<h3> This is h3 Heading </h3>

<h4> This is h4 Heading </h4>

<h5> This is h5 Heading </h5>

<h6> This is h6 Heading </h6>

<!doctype html>

<html>

<head>

<title>Sample Page</title>

</head>

<body>


```
<h1> This is h1 Heading </h1>
<h2> This is h2 Heading </h2>
<h3> This is h3 Heading </h3>
<h4> This is h4 Heading </h4>
<h5> This is h5 Heading </h5>
<h6> This is h6 Heading </h6>
```

```
</body>
</html>
```

cant we write bigger then that

Paragraph tag:

```
<p> This is my first paragraph</p>
```

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<title>Sample Page</title>
```

```
</head>
```

```
<body>
```

```
<h1> This is h1 Heading </h1>
```

```
<p> This is my first paragraph This is my first paragraphThis is my
first paragraphThis is my first paragraphThis is my first paragraphThis
is my first paragraphThis is my first paragraphThis is my first
```

paragraphThis is my first paragraphThis is my first paragraphThis is my first paragraphThis is my first paragraphThis is my first paragraphThis is my first paragraphThis is my first paragraphThis is my first paragraph</p>

<p>This is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraphThis is second paragraph</p>

</body>
</html>

can we increse decrease paragrph text font

case-1:

<p> This is First Line
 This is Second Line
 This is Third Line
 This is Fourth Line
</p>

Total data will come in a single line, because we are using only one paragraph tag.

case-2:

.....

<p> This is First Line</p>

<p> This is Second Line</p>

<p> This is Third Line</p>

<p> This is Fourth Line</p>

4 lines

case-3:

.....

<p> This is First Line</p> <p> This is Second Line</p><p> This is Third Line</p><p> This is Fourth Line</p>

4 lines

**Note: IN HTML idention is not important, but tags are important.
Blocking is also takes care by html tags only.**

**<p>This is First LineThis is First LineThis is First LineThis is First
LineThis is First LineThis is First LineThis is First LineThis is First
LineThis is First LineThis is First LineThis is First LineThis is First
LineThis is First LineThis is First LineThis is First LineThis is First**

**LineThis is First LineThis is First LineThis is First LineThis is First
Line</p>**

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8096969696**

**Session-1: <https://www.youtube.com/watch?v=bNjNNtpfLQw>
Session-2: <https://www.youtube.com/watch?v=l6eKrr8CJQA>
Session-3: <https://www.youtube.com/watch?v=md-Ztt-QUEk>**

<https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0Xulfozgly5Gs6j3j?usp=sharing>

**How many people are employees???
How many are waiting for job???
current academic students...**

How to plan our life???

How to plan career??

Poor will become more Poor--->Time Limit

Rich will become more Rich

Employee: $12 * 60 * 365 = 2,62,800$ Hours===>

IBM: 8 Lakhs hours time per day*60*365

17520000000===>How much money will be generated

50% for operations

1 L ---> 3Lakh employees

Microsoft 30L

Poor person always selling his time for rich people

Technical Labour

Rich people always buying others time....

100 employees 1 crore

200 employees 2 crores

10L --->

Ambani | Adhani--->24 hours only+others time

You --->24 hours only--->Sell your time

Wash your clothes

Wash clothes-->B.Tech/M.Tech/MCA

but how can we expect everything from others as we Are from middle class

Poor Dad and Rich Dad-->Robert

60 years

25 years

35 years * 365 days

12775 days on the earth

SHROLAN-MH2

S-->Speed

H--->Highest result work

R--->Reusable

O--->One time activitiy but continuous results

L--->Less efforts More results

A--->Automation

N--->New Activity for the next level

M--->Money

H--->Health

H-->Happiness

The Structure of HTML:

<html>

<head>

title

description

keywords

css and js files information

</head>

```
<body>
    original data which should be displayed to end user
</body>
</html>
```

comments:

```
<!-- this is html comment -->
```

heading tags:

```
<h1> .... to <h6>
```

paragraph tag

```
<p>
dskajfklsajlfkjsalkjflkdjsaljldsafkj
dskajfklsajlfkjsalkjflkdjsaljldsafkj
dskajfklsajlfkjsalkjflkdjsaljldsafkj
dskajfklsajlfkjsalkjflkdjsaljldsafkj
dskajfklsajlfkjsalkjflkdjsaljldsaf
</p>
```

Bold and Italic tags:

HTML: **** which is legacy

HTML 5: **** instead of ****

```
<!doctype html>
<html>
  <head>
    <title>Sample Page</title>
  </head>
  <body>
    <p> <strong>This is first line </strong></p>
    <p> This is <b>first line</b> </p>
  </body>
</html>
```

HTML: **<i>** is legacy

HTML 5: **** emphasis tag

```
<!doctype html>
<html>
  <head>
    <title>Sample Page</title>
  </head>
  <body>
    <p> <em><strong>This is first line </strong></em></p>
```

```
<p> <i>This is <b>first line</b></i> </p>
</body>
</html>
```

HTML Lists:

2 types of lists:

1. Ordered List(ol)
2. Unordered List(ul)

The available courses:

1. Python
2. Django
3. DataScience

The available courses:

- . Python
- . Django
- . DataScience

```
<ol>
```

```
<li> Python </li>
```

```
<li> Django </li>
```

```
<li> Data Science </li>
```

```
<li> DRF </li>
</ol>
```

```
<!doctype html>
<html>
  <head>
    <title>Sample Page</title>
  </head>
  <body>
    <h1> The Available Courses: </h1>
    <ul>
      <li> Python </li>
      <li> Django </li>
      <li> Data Science </li>
      <li> DRF </li>
    </ul>
  </body>
</html>
```

Nested Lists:

We can use list inside a list

```
<!doctype html>
```

```

<html>
<head>
  <title>Sample Page</title>
</head>
<body>
  <h1> THe Menu Items: </h1>
  <ol>
    <li> Food </li>
      <ul>
        <li> Chicken Biryani</li>
        <li> Mutton Biryani</li>
        <li> Fish Biryani</li>
      </ul>
    <li> Drinks </li>
      <ul>
        <li> KF</li>
        <li> KO</li>
        <li> RC</li>
      </ul>

    <li> Cigars </li>
  </ol>
</body>
</html>

```

Attributes:

We can pass attributes to the tags, which will provide some extra information.

Setting Image inside HTML:

```

```

src is the source of image

alt is alternate.

Note: We have to open the tag and we are not required to close, such type of tags are called self closing tags.

```
<img>
```

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<title>Sample Page</title>
```

```
</head>
```

```
<body>
```

```
<h1> The Walrus Operator is because of this animal (:=) </h1>
```

```

```

```
</body>
</html>
```

Note: Image from the google

```
<!doctype html>
<html>
<head>
  <title>Sample Page</title>
</head>
<body>
  <h1> Happy Colorful Holi Wishes from DURGASOFT!!! </h1>
  
</body>
</html>
```

```
<!doctype html>
<html>
<head>
  <title>Sample Page</title>
</head>
```

```
<body>
  <h1> Oye...Still you are Learning Python....</h1>
  
</body>
</html>
```

Creating Hyperlinks by using anchor tag:

```
<a> Click here to get more vidoes </a>
```

```
<!doctype html>
<html>
<head>
  <title>Sample Page</title>
</head>
<body>
  <h1> DURGASOFT</h1>
  <a
href="https://www.youtube.com/channel/UCbjozK_PYCTLEluFlrJ8UZ
g"> Click here to get more vidoes </a>

</body>
</html>
```

Mini Project:

```
<!doctype html>
<html lang="en">
<head>
  <title>Durga Bar and Restaurant</title>
</head>
<body>
  <h1> Welcome to Durga Bar and Restaurant</h1>
    
    <p>We are very specialized for Chilled Beer. We have virtual rain
environment with all facilities so that 100% guarantee for the Kick.
We have almost around 1000 brands like takela..We are very
specialized for Chilled Beer. We have virtual rain environment with all
facilities so that <strong>100% guarantee</strong> for the Kick. We
have almost around <strong>1000 brands</strong>like
takela..</p><p>We are very specialized for Chilled Beer. We have
virtual rain environment with all facilities so that 100% guarantee for
the Kick. We have almost around 1000 brands like takela..We are very
specialized for Chilled Beer. We have virtual rain environment with all
facilities so that 100% guarantee for the Kick. We have almost around
1000 brands like takela..</p>

  <h2>Our Brand Ambassodor is : Sunny Leone</h2>
  
```



```
<h3> Our offered brands are:</h3>
<ol>
  <li> King Fisher</li>
  <li> Knock Out</li>
  <li> Royal Stag</li>
  <li> 100 Pipers</li>
  <li> Foster</li>
  <li> Teachers Choice</li>
</ol>
<h1> Offer of the Day: <em>BUY ONE GET ONE FREE</em></h1>
<p> View Our Profile</p>
<a href="http://youtube.com"> Youtube</a>
<a href="http://facebook.com"> Facebook</a>
<a href="http://twitter.com"> Twitter</a>
</body>
</html>
```

sir if we click on that link directly it is going to that page directly.... but if we want to open that in new tab how to do that sir

Youtube
youtube.com/durgasoftware

sir how can I understand use p or h for adding some sentences

para
heading

Session-1: <https://www.youtube.com/watch?v=bNjNNtpfLQw>

Session-2: <https://www.youtube.com/watch?v=l6eKrr8CJQA>

Session-3: <https://www.youtube.com/watch?v=md-Ztt-QUEk>

Session-4: https://www.youtube.com/watch?v=kx9BIJ__EKY

<https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0XulfozglY5Gs6j3j?usp=sharing>

Comment: <!-- comment -->

header tags <h1> to <h6>

paragraph tag <p>

bold or

italic <i> or

lists ,

img

anchor text

Div and Span tags:

div means division

We can use <div> tag to group related tags into a single unit.

So that we can define styling for all these tags

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<link rel='stylesheet' href='style.css'/>
```

```
<title>Durga Bar and Restaurant</title>
```

```
</head>
```

```
<body>
```

```
<div>
```

```
<h1> UI Technologies are very important in web  
development </h1>
```

```
<h2> UI Technologies are very important in web  
development </h2>
```

```
<h3> UI Technologies are very important in web  
development </h3>
```

```
<h2> UI Technologies are very important in web  
development </h2>
```

```
<h1> UI Technologies are very important in web  
development </h1>
```

```
</div>
```

```
<h1> UI Technologies are very important in web development  
</h1>
```

```
<div>
    <h1> UI Technologies are very important in web
development </h1>
    <h2> UI Technologies are very important in web
development </h2>
    <h3> UI Technologies are very important in web
development </h3>
    <h2> UI Technologies are very important in web
development </h2>
    <h1> UI Technologies are very important in web
development </h1>
</div>
</body>
</html>
```

span tag:

life span

For inline styling : span tag

<h1> The 3 important viruses for the human being: Drinking, Gambling and Extra Curricular Activities.. These are very dangerous than corona also </h1>

Table Creation:

<table> to create table

<thead> to specify head row

<th> to specify column data in head row(column name)

<tr> to insert a row in the table

<td> to specify column data in the row/record.

<!doctype html>

<html>

<head>

<title>Sample HTML</title>

</head>

<body>

<h1> DURGASOFT Employees Data</h1>

<table border='2'>

<thead>

<th>ENO</th>

<th>ENAME</th>

<th>ESAL</th>

```

    <th>EADDR</th>
</thead>
<tr>
    <td>100</td>
    <td>Sunny</td>
    <td>1000</td>
    <td>Mumbai</td>
</tr>
<tr>
    <td>100</td>
    <td>Sunny</td>
    <td>1000</td>
    <td>Mumbai</td>
</tr>
<tr>
    <td>100</td>
    <td>Sunny</td>
    <td>1000</td>
    <td>Mumbai</td>
</tr>
<tr>
    <td>100</td>
    <td>Sunny</td>
    <td>1000</td>
    <td>Mumbai</td>
</tr>
<tr>
    <td>100</td>

```

```
        <td>Sunny</td>
        <td>1000</td>
        <td>Mumbai</td>
    </tr>
</table>
```

```
</body>
</html>
```

Oscar Awards Winners List for 2021:

S.NO | Winner | Winner Pic | Category | Country flag | View Profile

```
<!doctype html>
<html>
<head>
    <title>Sample HTML</title>
</head>
<body>
    <h1> Oscar Awards Winners List for 2021</h1>
    <table border='2'>
        <thead>
            <th>S.NO</th>
            <th>WINNER</th>
```

```

<th>WINNER PIC</th>
<th>CATEGORY</th>
<th>COUNTRY FLAG</th>
<th>VIEW PROFILE</th>
</thead>
<tr>
<td>1</td>
<td>Sunny</td>
<td><img src='sunny.jpg' alt='image not available'
width='300' height='200'></td>
<td>Best Actress</td>
<td><img
src='https://upload.wikimedia.org/wikipedia/en/4/41/Flag_of_India.
svg' width='300' height='200'></td>
<td><a href='http://youtube.com/durgasoftware'>View
Profile</a></td>
</tr>
<tr>
<td>2</td>
<td>Deepika</td>
<td><img src='Deepika.jpg' alt='image not available'
width='300' height='200'></td>
<td>Best Actress</td>
<td><img src='uk.png' width='300' height='200'></td>
<td><a href='http://youtube.com/durgasoftware'>View
Profile</a></td>
</tr>
<tr>

```



```

        <td>3</td>
        <td>Priya</td>
        <td><img src='priya.jpeg' alt='image not available'
width='300' height='200'></td>
        <td>Best Actress</td>
        <td><img src='us.png' width='300' height='200'></td>
        <td><a href='http://youtube.com/durgasoftware'>View
Profile</a></td>
    </tr>

</table>

```

```

</body>
</html>

```

<https://attendee.gotowebinar.com/register/5368769457292403467>

Creation of HTML Forms:

login form
registration form
feedback form

```
<form>
```

```
</form>
```

```
<!doctype html>
<html>
<head>
  <title>Sample HTML</title>
</head>
<body>
  <h1> Student Registration Form</h1>

  <form action='result.html' method='GET'>
    Name: <input type='text' name='', value=''>
    <input type='submit' value='login'>
  </form>

</body>
</html>
```

Within the form we have to collect end user input.
To collect end user input, we have to use <input> tag.

```
<input type='text' name='' value='' >
```

type attribute represents the type of input

text
email
password
radio
checkbox
color

username contain number no sir

```
<input type='text' name="" value="" >  
<input type='password' name="" value="" >  
<input type='email' name="" value="" >  
<input type='color' name="" value="" >  
<input type='radio' name="" value="" >  
<input type='checkbox' name="" value="" >
```

```
<input type='text' name="" value='Enter Name' >
```

Name attribute represents the name of input tag. By using this in the next target page we can access its value.

value attribute represents default value will be displayed in the form.

type

name

value--->To provide default value.

placeholder --->attribute is just to provide hint but not default value.

Recommended is placeholder

required attribute:

Name2 <input type='text' name='name2' placeholder='Enter Name2' required>

Creation of Labels for HTML Elements:

<label for="name"> Enter Name:</label>

<input id='name' type='text' name='uname' value='Enter Name'>

The value of for attribute should be same as id attribute value.

Implementing Radio Button:

<h3> Are You Married:</h3>

<label for="yes">Yes</label>

<input id='yes' type='radio' name='married' value="">

<label for="no">No</label>

**<input id='no' type='radio' name='married' value="">
**

eg:

<h3> Gender</h3>

<label for="m">Male</label>

<input id='m' type='radio' name='gender' value='male'>

<label for="fm">Female</label>

**<input id='fm' type='radio' name='gender'
value='female'>**

<label for="t">Transgender</label>

**<input id='t' type='radio' name='gender'
value='transgender'>
**

sir my question is that when we develop any web application so mostly html webpage write in edit plus.

sir why name is written as => Name: and why not Password: ??

Session-1: <https://www.youtube.com/watch?v=bNjNNtpfLQw>

Session-2: <https://www.youtube.com/watch?v=l6eKrr8CJQA>

Session-3: <https://www.youtube.com/watch?v=md-Ztt-QUEk>

Session-4: https://www.youtube.com/watch?v=kx9BIJ__EKY

<https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0Xulfozgly5Gs6j3j?usp=sharing>

COMMENTS

Header tags

paragraph

div

span

ol and ul

tables

images

anchor tag

forms

```
<form action='second.html'>
```

```
  <input type='text' ....>
```

```
  <input type='email' ....>
```

```
  <input type='password' ....>
```

```
  <input type='color' ....>
```

```
  <input type='radio' ....>
```

```
  <input type='submit' ....>
```

```
</form>
```

```
<input type='text' name='byusingthisname' value='defaultvalue'>
```

value--->To provide default value

placeholder--->just to provide guidelines to the end user

required

<label for='name'>Name</label>

<input id='name' type='text' name='username' value=''>

page inspection

css styling also

file:///D:/django26/result.html?username=Sunny

In GET method, the data will be appended to the url and send to the next page.

In Post method, the data will be encapsulated inside request body and not visible in the url.

file:///D:/django26/result.html

How to place multiple submit buttons in the same form?

file:///D:/django26/result.html?username=Sunny&action=ADD

file:///D:/django26/result.html?username=Sunny&action=REMOVE

file:///D:/django26/result.html?username=Sunny&action=UPDATE

<!doctype html>

```

<html>
<head>
    <title>Sample HTML</title>
</head>
<body>
    <h1> Student Registration Form</h1>

    <form action='result.html' method='GET'>
        Name: <input type='text' name='username' value=''><br>
            <input type='submit' name='action' value='ADD'><br>
            <input type='submit' name='action'
value='REMOVE'><br>
            <input type='submit' name='action' value='UPDATE'>
    </form>

</body>
</html>

```

```

action = request.getParameter('action')
if action == 'ADD':
    perform add functionality
elif action == 'REMOVE':
    perform remove functionality
elif action == 'UPDATE':
    perform update functionality

```


<input type='text'>
<input type='email'>
<input type='password'>
<input type='color'>
<input type='radio'>
<input type='submit'>

Gender: Male Female

Online exam webapplication:

Q1. Question (Select only one option)

- A. sdklfjdsa**
- B. klsadjflksad**
- C. ksdjflkjsa**
- D. sdlkf;lsk**

**Several options are available but we have to select only one option:
radio button**

<h1> Q1. This is Sample Question</h1>

**<input type='radio' name='q1' value='a'> A
**

```
<input type='radio' name='q1' value='b'> B<br>
<input type='radio' name='q1' value='c'> C<br>
<input type='radio' name='q1' value='c'> D<br>
<input type='submit' value='Submit Exam'>
```

file:///D:/django26/result.html?q1=a&q1=c

file:///D:/django26/result.html?q1=a&q1=c&q1=d

```
<!doctype html>
<html>
<head>
  <title>Sample HTML</title>
</head>
<body>

  <form action='result.html' method='GET'>
    <h1> Q1. This is Sample Question (Choose all valid options)</h1>
    <input type='checkbox' name='q1' value='a'> A<br>
    <input type='checkbox' name='q1' value='b'> B<br>
    <input type='checkbox' name='q1' value='c'> C<br>
    <input type='checkbox' name='q1' value='d'> D<br>
    <input type='submit' value='Submit Exam'>
  </form>

</body>
</html>
```

```

<!doctype html>
<html>
  <head>
    <title>Sample HTML</title>
  </head>
  <body>

    <form action='result.html' method='GET'>
      <h1> Choose Your Known Languages</h1>
      <input type='checkbox' name='language' value='eng'>
English
      <input type='checkbox' name='language' value='tel'>
Telugu
      <input type='checkbox' name='language' value='tamil'>
Tamil
      <input type='checkbox' name='language' value='hindi'>
Hindi<br>

      <input type='submit' value='Submit'>
    </form>

  </body>
</html>
file:///D:/django26/result.html?language=eng&language=tel&language=tamil

```

`<input type='checkbox' name='language' value='hindi' checked>`
Hindi

Choose Your Country
Choose Your State

How to implement Dropdown box/select box:

file:///D:/django26/result.html?state=ka
file:///D:/django26/result.html?state=mh

file:///D:/django26/result.html?state=ap&state=up&state=mh

```
<h1> Choose Your State</h1>
  <select name='state' multiple>
    <option value='ap'>Andhra Pradesh</option>
    <option value='ka'>Karnataka</option>
    <option value='ts'>Telangana</option>
    <option value='tn'>Tamil Nadu</option>
    <option value='up'>Uttara Pradesh</option>
    <option value='mp'>Madya Pradesh</option>
    <option value='mh'>Maharastra</option>
  </select>
```

Radio vs checkbox vs select box:

If few options are available and we have to choose only one option:
radio

If few options are available and we can select any number of options:
checkbox

If very huge number of options are available and we can select any
number of options: dropdown box/select box

Out of 5 options you have to choose only 3 then what we have to do
sir

expecting some functionality--->js

textarea:

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<title>Sample HTML</title>
```

```
</head>
```

```
<body>
```

```
<form action='result.html' method='GET'>
```

```
<h1> Enter Feedback</h1>
```

```
<textarea name='feedback' rows='8' cols='80'>
```

```
</textarea><br>
```

```
<input type='submit' value='Submit'>  
</form>
```

```
</body>  
</html>
```

if form have textbox, radio button , dropdown , checkbox, then on submit click , all the data go through the URL (in case of GET method)

sir I want to hide data in URL leak in url shall i use POST method

GET vs POST:

If we want to get information from the server-->GET

Download movie but we have to provide movie name-->get

If we want to post information to the server--->

upload student information

upload student feedback ----->POST

URL query string--->4000+ characters

sir why not use post for everything?

<https://www.youtube.com/watch?v=E-Y5CeeYJxU>

<https://www.youtube.com/>

atom.io

<https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0XulfozglY5Gs6j3j?usp=sharing>

Durgamatrimonial app:

Styling is missing

Functionality is missing- Backend is required

ctrl+shift+d--->print current line again

sir in dropdown i want to keep default value as "select any one"

Good,if I select 2 what boss like...is possible

<file:///D:/django26/durgamatrimonial1/thankyou.html?username=Durga&email=info%40subjectroom.com&age=60&dating=Yes&expectations=tamanna&nom=3&alchol=yes&name=No+special+preferences>

GET Method(Default Method)

POST

file:///D:/django26/durgamatrimonial1/thankyou.html

sir is there anyway we can verify if provided email id is valid or not here ?

name attribute is same for the all tags so how can we differentiate in backend which name attribute is belongs to tag sir...?

name attribute is the same but value is different

durgasoftonlinetraining@gmail.com

8885252627

8096969696

<https://www.youtube.com/playlist?list=PLd3UqWTnYXOkA-bkmcwkbY1iJkcU4aM1r>

h1,h2,h3...,h6

p

div

span

or,ul

text

submit

radio

password

email

color

checkbox

textarea

b or strong

i or em

dropdown box

table

**
**

HTML Validations:

1. email

2. required

3. min and max length

4. only digits are allowed

5. only lower case alphabet symbols are allowed

etc

Case-1: The value should contain minimum 5 characters and maximum 10 characters.

```
<input type="text" name="val" value="">
```

```
<input type="text" name="val" value="" pattern="">
```

In Regular Expressions:

[0-9] --->Any digit (\d)

[a-z] --->Only lower case alphabet symbol

[A-Z] --->Only upper case alphabet symbol

[a-zA-Z] --->Any alphabet symbol

[a-zA-Z0-9] --->Any alphanumeric character (\w)

. --->Any character

[abc] --->either a or b or c

a* --->Any number of a's including zero number

a+ --->Atleast one a

Case-1: The value should contain minimum 5 characters and maximum 10 characters.

.{5,10}

<input type="text" name="val" value="" pattern="" title="msg">

Case-2: The value should contain only digits, exactly 10 digits only allowed.

pattern='[0-9]{10}'

Case-3: The value should contain only digits, exactly 10 digits only allowed.

But first digit should be either 6 or 7 or 8 or 9.

pattern='[6-9][0-9]{9}'

[6-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9]

[6-9][0-9]{9}

9898989897

6786786789

4564564567

case-4:

password atleast 8 characters must be required.

**The first character should be upper case alphabet symbol
remaining any alpha numeric.**

[A-Z][a-zA-Z0-9]{7}[a-zA-Z0-9]*

sir we can put first Letter Uppercase and remaing 2 letter is small and last character is special symbol

[A-Z][a-z]{2}[^a-zA-Z0-9]

[abc]--->either a or b or c

[a-zA-Z0-9]--->any alphanumeric character

[^abc] --->Except a and b and c any other character

[^a-zA-Z0-9] --->Any character except alpha numeric character

[A-Z][a-z]{2}[^a-zA-Z0-9]

mailid validation regular expression

How to include spaces in HTML:

- 1. to add a single space**
- 2. &ensp to add 2 spaces**
- 3. &emsp to add 4 spaces**

**4.
 to add a line break**

5. <hr>

Basic Idea about HTML

CSS(Cascading Style Sheets):

The main objective of CSS is to add styles to HTML.

CSS describes how html elements are displayed on web page.

Styling includes colors,fonts,size,borders etc

sir with html we can create only structure of the webpage but by the css we can develop beautiful webpage i am correct sir

Atom shortcuts:

ctrl+shift+d --->Duplicate current line

ctrl+shift+c --->To copy current page full path

ctrl+/ --->to comment/uncomment

File-->Settings-->Editor--->Softwrap =====>word wrapping

Various Ways to define style for HTML Elements:

3 ways

1. In Line

2. By using style tag

3. By using external CSS file

1. In Line:

```
<h1 style="color:red"> This is for CSS Demo</h1>
```

2. By using style tag:

```
<style>
```

```
<head>
```

```
  <style>
```

```
  h1{
```

```
    color:red;
```

```
  }
```

```
  </style>
```

```
  <meta charset="utf-8">
```

```
  <title>Sample Page</title>
```

```
</head>
```

```
<!DOCTYPE html>
```

```
<html lang="en" dir="ltr">
```

```
  <head>
```

```
    <style>
```

```
    h1{
```

```
      color:blue;
```

```
    }
```

```
    </style>
```

```
    <meta charset="utf-8">
```

```

<title>Sample Page</title>
</head>
<body>

<h1> This is for CSS Demo</h1>
<h1> This is for CSS Demo</h1>
<h1> This is for CSS Demo</h1>
<h1> This is for CSS Demo</h1>

</body>
</html>

```

3. By using external CSS file:

```

<link rel='stylesheet' href='style2.css'>

<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title></title>
    <link rel="stylesheet" href="style2.css">
  </head>
  <body>
    <h1> This is for CSS Demo</h1>

```

```
<h1> This is for CSS Demo</h1>
<h1> This is for CSS Demo</h1>
<h1> This is for CSS Demo</h1>
</body>
</html>
```

sir...I want different color for both page..then how we can show.?

Basic Structure of CSS File:

```
tagname{
  property:value;
}
```

```
h1{
  color:orange;
}
```

```
<head>
  <link rel='stylesheet' href='style.css'>
</head>
```

Demo Application-1:

<https://www.youtube.com/playlist?list=PLd3UqWTnYXOkA-bkmcwkbY1iJkcU4aM1r>

Various Possible ways to specify color:

1.

```
h1{  
  color:red;  
}
```

2.

color:rgb(0,0,0) ---->Black
color:rgb(255,255,255) ---->White

r---->Red

g-->Green

B-->Blue

Universal colors

The allowed range of values: 0 to 255

3. 6-digit hexa decimal code

color: #f44e42

4. rgba

`color:rgba(255,255,255)`

a means alpha--->Transparency
0.0 to 1.0

0.0 means full light
1.0 means full dark

Red

Setting background:

`background:`

`background-color:`

color vs background-color

`color-->text color`

`background color--->`

How to set background image:

`body{`

```
background:
url(https://www.remove.bg/images/remove_image_background.jpg);
background-repeat: no-repeat;
background-size: cover;
}
```

Note:

<!-- this is HTML comment -->

/*

This is css comment

***/**

How to set Border:

```
img{
border-color: red;
border-width: 20px;
border-style:groove;
}
```

shortcut way:

```
img{
```

```
    border: 30px dashed blue;  
}
```

```
border: red 20px groove;
```

Here order is not important

Basic CSS Selectors:

1. Element Selectors
2. ID Selectors
3. Class Selectors

1. Element Selectors:

Selects all instances of given element and apply the style.

```
h1{  
    color: red;  
}
```

2. ID Selectors:

Selects an element with given Id. But within HTML, ID should be unique.

```
#specialh1{  
  color:blue;  
}
```

3. class selectors:

selects all elements with the given class and apply styling.

```
.special{  
  color: red;  
}
```

For every instance of specified tag==>Element selectors

For only one tag instance==>ID selectors

For multiple tags (may be different tags) -->class selectors

100 h1 tags

for all 100 h1 tags red color==>element selectors

only one h1 tag should be displayed in blue color==>id selector

just 15 h1 tags should be displayed in orange color==>class selector

```
element{  
  property: value;  
}
```

```
#element{  
  property: value;  
}
```

```
.element{  
  property: value;  
}
```

```
<!DOCTYPE html>  
<html lang="en" dir="ltr">  
  <head>  
    <meta charset="utf-8">  
    <title>Sample Page</title>  
    <link rel="stylesheet" href="s1.css">  
  </head>  
  <body>  
    <h1>The Importance of CSS</h1>  
    <h1 class='special'>The Importance of CSS</h1>  
    <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do  
    eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim  
    ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut  
    aliquip ex ea commodo consequat. Duis aute irure dolor in  
    reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla  
    pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa  
    qui officia deserunt mollit anim id est laborum.</p>
```

<h1>The Importance of CSS</h1>

<h1 id="specialh1">The Importance of CSS</h1>

**<p class='special'>Lorem ipsum dolor sit amet, consectetur
adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore
magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute
irure dolor in reprehenderit in voluptate velit esse cillum dolore eu
fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident,
sunt in culpa qui officia deserunt mollit anim id est laborum.</p>**

<h1 class='special'>The Importance of CSS</h1>

</body>

</html>

Basic CSS Selectors:

- 1. Element selectors---->All instances**
- 2. ID selectors--->Only for one instance(unique)**
- 3. class selectors--->Multiple instances but not all**

**Select multiple instances of div tag(but not all div tags) and apply style
class selectors**

<div>

to group tags

to group tags

to group tags

to group tags
</div>

Advanced CSS Selectors:

1. * selector
2. Descendant selector
3. Adjacent selector
4. Attribute Selector
5. nth of type selector

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>Sample Page</title>
    <link rel="stylesheet" href="s1.css">
  </head>
  <body>
    <h1>Advanced Selectors Demo</h1>
    <a href="http://google.com">Click Here to goto Google</a>
    <ul>
      <li>CAT</li>
      <li>RAT</li>
```



```
<li>ANT</li>
<li>DOG</li>
<li>TIGER</li>
</ul>
<h4>List of Movies:</h4>
<ul>
  <li>Bahubali</li>
  <li>VakeelSab</li>
  <li>KGF-2</li>
  <li>F2</li>
  <li>F3</li>
</ul>
<h4>List of top websites:</h4>
<ul>
  <li> <a href="http://amazon.com">AMAZON</a> </li>
  <li> <a href="http://flipkart.com">FLIPKART</a> </li>
  <li> <a href="http://paytm.com">PAYTM</a> </li>
  <li> <a href="http://durgasoftonline.com">DURGASOFT</a> </li>
</ul>
</body>
</html>
```

1. * selector:

*** means everything**

This style is applicable for every tag in the web page

```
*{  
  color:blue;  
  background: yellow;  
}
```

2. Descendant Selector:

sub level-->child tag

```
p span{  
  color: red;  
}
```

Style is applicable only for span tags which are present inside p tag.

```
li a{  
  color: red;  
}
```

Style is applicable only for anchor tags which are present inside li tag.

```
body ul li a{
```

```
color:blue;
background: yellow;
}
```

child need not be immediate child.

3. Adjacent Selectors:

```
h1+a{
color:blue;
background:yellow;
}
```

select all anchor tags which are adjacent to h1 and apply styling.

```
div+p{
color:red;
}
```

apply style for p tags which are adjacent to div tag.

```
h4+ul{
color:white;
background:red;
}
```

```
h1+h4{  
}
```

```
div ul+a{  
}
```

select all anchor tags which are adjacent to ul and that ul present inside div tag.

*** selector**

Descendant selector

Adjacent selectors

1. Problems(ofcourse viruses) are very common. We have to take much care and we have to drive our life very carefully. Feeling sad is time waste activity.

2. Everyone has his own problems. There is no person in this world without any problem.

3. Every problem is best guru for us. It will provide n number of lessons.

4. Happiness in our life is the most important thing, we should not miss any time this one.

5. Identify small small situations to celebrate.

Corona how worst it is in our life???

10 (good points) lessons from corona???

problems, sadness, angryness, worry, lack of money, job related problems, family problems ---->Your own feelings

Happiness, pleasantness, smile --->

CSS Selectors:

To style/makeup html elements(h1,p,div,...)

We have to select html elements and we have to apply styling. To select html elements we have to use css selectors.

Basic CSS Selectors:

1. element selectors

```
h1{  
  color:red;
```

```
}
```

2. id selectors:

```
#special{  
    color:red;  
}
```

3. class selectors:

```
.special{  
    color: red;  
}
```

Advanced CSS Selectors:

1. * selector

```
{  
    color:red;  
}
```

2. Descendant Selectors:

```
p c{  
    color: red;  
}
```

```
ul a{
```

```
color: red;
}
```

3. Adjacent Selectors:

```
ul+a{
  color: red;
}
```

4. Attribute selector:

We can define style based on attribute also.

```
input[type='password']{
  background:yellow;
}
```

```
<input type='text'
<input type='password'
<input type='email'
<input type='text'
```

```
input[type='password']{
  background: yellow;
}
```

```
a[href="http://flipkart.com"]{
  background: red;
}
```

5. nth of type selectors:

```
<ul>
  <li>XXX</li>
  <li>XXX</li>
  <li>XXX</li>
  <li>XXX</li>
  <li>XXX</li>
</ul>
```

```
<ul>
  <li>XXX</li>
  <li>XXX</li>
  <li>XXX</li>
  <li>XXX</li>
  <li>XXX</li>
</ul>
```

```
ul:nth-of-type(2){
  background: red;
  color:white;
```



```
}
```

```
li:nth-of-type(even){  
  background: red;  
  color:white;  
}
```

in li for first and second one color and for third and fourth another color like that we will do in css sir

Basic: 3 (element,id,class)

Advanced: 5(*,descendant,adjacent,attribute,nth of type)

CSS Inheritance:

What every styles parent tag has, by default will be available for child tags also.

CSS Specificity:

If multiple styles are available for element, then the most specific style will be considered.

parent-->style1

child--->style2

sir if we want for every list different color then what sir

Assignment-1:

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>CSS Assignment</title>
  </head>
  <body>
    <h1>This is h1 Data</h1>
    <h2>This is h2 tag</h2>
    <h2>This is h2 tag</h2>
    <h2>This is h2 tag</h2>
    <h3>The important Brands are:</h3>
    <ul>
```

`<li>King Fisher</li>`

`<li>Royal Challenge</li>`

`<li>Foster</li>`

`</ul>`

`<p class="hello">Lorem ipsum dolor sit amet, consectetur
adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore
magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute
irure dolor in reprehenderit in voluptate velit esse cillum dolore eu
fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident,
sunt in culpa qui officia deserunt mollit anim id est laborum.</p>`

`<p id="special">Lorem ipsum dolor sit amet, consectetur adipiscing
elit, sed do eiusmod tempor incididunt ut labore et dolore magna
aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco
laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa
qui officia deserunt mollit anim id est laborum.</p>`

`<label for="name">Name</label>`

`<input type="text" id="name" name="username"
value="">

`

`<label for="pwd">Password</label>`

`<input id="pwd" type="password" name="password"
value="">

`

`<a href="http://www.gmail.com">Gmail</a>`

`<a href="http://www.youtube.com">Youtube</a>`

`<a href="http://www.facebook.com">Facebook</a>`

`<h3>Select Your Favourite Heroine:</h3>`

`<ul>`

`<li>Sunny: <input type="checkbox" name="" value="" > </li>`

`<li>Mallika: <input type="checkbox" name="" value="" > </li>`

`<li>Veena Malik: <input type="checkbox" name="" value="" >`

`</li>`

`</ul>`

`</body>`

`</html>`

Cases:

1. Set body tag background color with #bdc646

body{

background

2. Set h1 text color with #9b79b8

3. Make all h2 tags with yellow color

4. Make all elements with blue color, but consider hexa code

5. Change background for every paragraph with white color

6. Make all inputs have a 5px border with red color

**7. Set blue background with class attribute hello
and text color as white**

8. Give 10px orange solid border for id attribute "special"

9. Make only inputs with type 'text' have a green background

10. Make all "checked" check boxes have a left margin of 50px
11. Make the <label> elements all UPPERCASE without changing in html
12. Make the first letter of the element with id "special" with green color and 100 px font-size
13. Make <h1> element color change to white when hovered and give 10px solid red border
14. Make all <a> elements that have been visited with yellow color

```
/* 1. Set body tag background color with #bdc646 */
```

```
body{  
    background: #bdc646;  
}
```

```
/* 2. Set h1 text color with #9b79b8 */
```

```
h1{  
    color:#9b79b8;  
}
```

```
/* 3. Make all h2 tags with yellow color */
```

```
h2{  
    color:yellow;  
}
```

```
/* 4. Make all <li> elements with blue color,but consider hexa code */
```

```
li{  
    color:#2515d1;  
}
```

```
/* 5. Change background for every paragraph with white color */  
p{  
    background: white;  
}
```

```
/* 6. Make all inputs have a 5px border with red color */  
input{  
    border: 5px red groove;  
}
```

```
/* 7. Set blue background with class attribute hello  
and text color as white */  
.hello{  
    background: blue;  
    color:white;  
}
```

```
/* 8. Give 10px orange solid border for id attribute "special" */  
#special{  
    border: 10px orange solid;  
}
```

```
/* 9. Make only inputs with type 'text' have a green background */  
input[type='text']{  
    background: green;  
}
```

/* 10. Make all "checked" check boxes have a left margin of 50px */

```
input:checked{  
  margin-left: 250px;
```

```
}
```

/* 11. Make the <label> elements all UPPERCASE without changing in html */

```
label{  
  text-transform: uppercase;  
}
```

/* 12. Make the first letter of the element with id "special" with green color and 100 px font-size */

```
#special:first-letter{  
  color:green;  
  font-size: 50px;  
}
```

/* 13. Make <h1> element color change to white when hovered and give 10px solid red border */

```
h1:hover{  
  color:white;  
  border: 10px solid red;  
  
}
```

```
/* 14. Make all <a> elements that have been visited with yellow color  
*/  
a:visited{  
    color:yellow;  
}
```

**[https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0Xulfozgl
y5Gs6j3j?usp=sharing](https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0Xulfozgl
y5Gs6j3j?usp=sharing)**

**sir, can you show the background color for every 2nd row in tag
once again.**

CSS Selectors:

1. element selector

```
h1{  
    color: red;  
}
```

2. ID selectors

```
#specialid{  
    color: red;  
}
```

3. class selector


```
.classname{  
  color: red;  
}
```

4. * selector

```
{  
  color: red;  
}
```

5. Descendant Selector:

```
div p {  
  color: red;  
}
```

6. Adjacent Selectors

```
div+p{  
}
```

7. Attribute selectors:

```
input[type='password']{  
  
}
```

8. nth of type selector:

```
p:nth-of-type(3){  
  color:red;  
}
```

CSS Inheritance:

CSS Specificity:

```
<body>  
  <p>  
  <div>  
    <p>  
    <p>  
    <p>  
  </div>  
  <div>  
    <p>  
    <p>  
    <p>  
  </div>  
  <div>  
    <p>  
    <p>  
    <p>  
  </div>  
  <p>
```

</body>

Assignment2.html:

<!DOCTYPE html>

<html lang="en" dir="ltr">

<head>

<meta charset="utf-8">

<link rel="stylesheet" href="assignment2.css">

<title></title>

</head>

<body>

<p>This Para is outside of div. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. </p>

<div>

<p>This Para is inside of first div. em tag inside first div in voluptate velit esse cillum dolore eu fugiat nulla pariatur. </p>

**<p>This Para is inside of first div. em tag inside first
div in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>**

**<p>This Para is inside of first div. em tag inside first
div in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>
</div>**

**<div>
<p>This Para is inside of Second div. em tag inside Second
div in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>**

**<p>This Para is inside of Second div. em tag inside Second
div in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>**

**<p>This Para is inside of Second div. em tag inside Second
div in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>
</div>**

**<div>
<p>This Para is inside of Third div. em tag inside Third
div in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>**

```
<p>This Para is inside of Third div. <em> em tag inside Third
div</em> in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>
```

```
<p>This Para is inside of Third div. <em> em tag inside Third
div</em> in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
</p>
```

```
</div>
```

```
<p>This Para is outside of div. <em> Duis aute irure dolor in
reprehenderit</em> in voluptate velit esse cillum dolore eu fugiat
nulla pariatur. </p>
```

```
</body>
```

```
</html>
```

assignment2.css:

```
/* Case-1: Make all <p>s That are nested inside div with 25px font-size
*/
```

```
div p{
  font-size: 25px;
}
```

```
/* Case-2: Give all <p>s in side 3rd div blue background and text color
is white */
```

```
div:nth-of-type(3) p{
  background: blue;
  color: white;
```

```
}
```

```
/* Case-3: Give 2nd <p> inside 3rd div, 5px red solid border */
```

```
div:nth-of-type(3) p:nth-of-type(2){
```

```
  border: 5px solid red;
```

```
}
```

```
/* Case-4: Make <em> tag in the 2nd <div> with white color 30 px  
font-size and green background */
```

```
div:nth-of-type(2) em{
```

```
  color: white;
```

```
  font-size: 30px;
```

```
  background: green;
```

```
}
```

Fonts and Text in css:

1. font-family

2. font-size

3. font-weight

4. line-height

5. text-align

6. text-decoration

1. font-family:

cssfontstack.com

2. font-size:

font-size: 2.0em;

2.0 em units-->Dynamic font size (relative font size)

body

p

span

p{

font-size: 30px;

}

span{

font-size: 0.5em;

}

3. font-weight:

100 to 900

100 means light

900 means too much bold

4. line-height:

The space between 2 lines

5. text-align:

6. text-decoration:

durga

durga

durga

cssfontstack.com

How to use custom fonts in css:

1. copy link tags and paste inside head part of html

<link rel="preconnect" href="https://fonts.gstatic.com">

<link

href="https://fonts.googleapis.com/css2?family=Bungee+Shade&display=swap" rel="stylesheet">

2. add css property

h1{

font-family: 'Bungee Shade', cursive;

}

The Box Model:

Every HTML Element is a represented as a rectangular box.

Each box has 4 edges.

- 1. Content**
- 2. Padding**
- 3. Border**
- 4. Margin**

padding: 20px 40px 60px 80px;

top: 20px

Right: 40px

Bottom: 60Px

Left: 80Px

vibgyor Application:

v-->violet

html

```
<!DOCTYPE html>
```

```
<html lang="en" dir="ltr">
```

```
<head>
```

```
<meta charset="utf-8">
```

```
<title>Vibgyor Implementation</title>
```

```
<link rel="stylesheet" href="vibgyor.css">
```

```
<link rel="preconnect" href="https://fonts.gstatic.com">
```

```
<link
```

```
href="https://fonts.googleapis.com/css2?family=Original+Surfer&dis  
play=swap" rel="stylesheet">
```

```
</head>
```

```
<body>
```

```
<h1>VIBGYOR Implementation</h1>
```

```
<table>
```

```
<tr>
```

```
<td id='one'></td>
```

```
<td id='two'></td>
```

```
<td id='three'></td>
```

```
<td id='four'></td>
```

```
<td id='five'></td>
```

```
<td id='six'></td>
```

```
<td id='seven'></td>
```

```
</tr>
```

```
</table>
```

```
</body>
```

```
</html>
```

CSS:

```
h1{  
  text-align: center;  
  font-family: 'Original Surfer', cursive;  
}  
table{  
  margin:auto;  
}  
td{  
  height: 300px;  
  width: 150px;  
  border:6px solid white;  
}  
#one{  
  background: #9400D3;  
}  
#two{  
  background: #4B0082;  
}  
#three{  
  background: #0000FF;  
}  
#four{  
  background: #00FF00;  
}
```

```
#five{
  background: #FFFF00;
}
#six{
  background: #FF7F00;
}
#seven{
  background: #FF0000;
}
```

Chess Board Implementation:

html:

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>Chess Board Implementation</title>
    <link rel="stylesheet" href="chess.css">
  </head>
  <body>
    <h1>Chess Board Implementation</h1>
    <table>
      <tr>
        <td class='white'></td>
        <td class='black'></td>
```

```
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
</tr>
<tr>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
</tr>
<tr>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
</tr>
<tr>
```

```

<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
</tr>
<tr>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
</tr>
<tr>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>
<td class='black'></td>
<td class='white'></td>

```

```

</tr>
<tr>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
</tr>
<tr>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
  <td class='black'></td>
  <td class='white'></td>
</tr>
</table>

</body>
</html>

```

CSS:


```

---
h1{
  text-align: center;
}
td{
  height: 60px;
  width: 60px;
}
.white{
  background: white;
}
.black{
  background: black;
}
table{
  border: 10px solid black;
  margin: auto;
}

```

Application-3: Tic-Tac-Toe Implementation:

```

-----
html:
----
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>Tic Tac Toe Game</title>

```

```

<link rel="stylesheet" href="ttt.css">
<link rel="preconnect" href="https://fonts.gstatic.com">
<link
href="https://fonts.googleapis.com/css2?family=Coda+Caption:wght
@800&display=swap" rel="stylesheet">
</head>
<body>
<h1>Tic Tac Toe Game</h1>
<table>
<tr>
<td></td>
<td class='vertical'></td>
<td></td>
</tr>
<tr>
<td class='horizontal'></td>
<td class='vertical horizontal'></td>
<td class='horizontal'></td>
</tr>
<tr>
<td></td>
<td class='vertical'></td>
<td></td>
</tr>
</table>

</body>
</html>

```

css:

```
td{  
  height: 100px;  
  width: 100px;  
}  
.vertical{  
  border-left: 5px solid blue;  
  border-right: 5px solid blue;  
}  
.horizontal{  
  border-top: 5px solid blue;  
  border-bottom: 5px solid blue;  
}  
table{  
  margin:auto;  
}  
h1{  
  text-align: center;  
  margin-bottom: 50px;  
  font-family: 'Coda Caption', sans-serif;  
}  
td:hover{  
  background: yellow;  
}
```

web applications--->Python Based Framework (Django)

Durga's Favourite Photo Album

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>Durga's Favourite Photo Album</title>
    <link rel="stylesheet" href="dpalbum.css">
    <link rel="preconnect" href="https://fonts.gstatic.com">
  <link
href="https://fonts.googleapis.com/css2?family=New+Tegomin&displ
ay=swap" rel="stylesheet">
  </head>
  <body>
    <p>Durga's Favourite Photo Gallery</p>
    
    
    
    
    
    
    
    
    
    
```

```


</body>
</html>
```

css:

```
img{
  height: 300px;
  width: 20%;
  float:left;
  margin: 2.5%;
}
/* img:hover{
  border: 3px groove red;
} */
p{
  text-align: center;
  font-size: 25px;
  font-weight: 900;
  font-family: 'New Tegomin', serif;
  border-bottom: 2px solid red;
  padding-bottom: 20px;
}
```

<https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0XulfozglY5Gs6j3j?usp=sharing>

1. Vibgyor
2. Chess
3. TTT
4. Photo Album
5. Durga Gambling Website
6. Blog app

5. Durga Gambling Website:

gambling.html:

<!DOCTYPE html>

<html lang="en" dir="ltr">

```
<head>
  <meta charset="utf-8">
  <title>DURGA GAMBLING GROUP</title>
  <link rel="stylesheet" href="gambling2.css">
  <link rel="preconnect" href="https://fonts.gstatic.com">
<link
href="https://fonts.googleapis.com/css2?family=Krona+One&display=swap" rel="stylesheet">
</head>
<body>
  <h1>DURGA GAMBLING GROUP</h1>
  <h2>India's No.1 Gambling Group</h2>
  <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do
eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim
ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut
aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa
qui officia deserunt mollit anim id est laborum.</p>
  <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do
eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim
ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut
aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa
qui officia deserunt mollit anim id est laborum.</p>
  <h2>Are You Interested to Join This Group:</h2>
  <form action='thankyou2.html' method='post'>
```

```

<label for='name'> Enter Name:</label><br>
<input id='name' type='text' name='name' value=''><br>
<label for='mail'> Enter Mail Id:</label><br>
<input id='mail' type='email' name='mail' value=''><br>
<label for='mno'> Enter Mobile Number:</label><br>
<input id='mno' type='text' name='mno' value=''><br>
<input id='button' type="submit" name="" value="Join Us">

</form>
</body>
</html>

```

thankyou.html:

```

<!DOCTYPE html>
<html lang="en" dir="ltr">
<head>
  <meta charset="utf-8">
  <title>Thank You Page</title>
  <link rel="stylesheet" href="gambling2.css">
  <link rel="preconnect" href="https://fonts.gstatic.com">
  <link
href="https://fonts.googleapis.com/css2?family=Krona+One&display
=swap" rel="stylesheet">
</head>
<body>

```


<h2>Thanks for your interest on Gambling</h2>

<h3>Our Experts will contact you soon !!!!</h3>

</body>

</html>

css file:

body{

background: #011627;

color:#fdffc;

text-align: center;

border: 40px solid #2EC486;

border-top: 0px;

margin: 0px;

font-family: 'Krona One', sans-serif;

}

p{

padding-left: 20%;

padding-right: 20%;

padding-top: 5%;

}

form{

font-size: 1.5em;

margin: 20px;

}

#button{

```
height: 70px;
width: 140px;
margin: 10px;
font-size: 1.5em;
background: orange;
border-radius: 25px;
}
```

sir please clarify ? if we provide 2 links .. then the first link will be overridden right ? then how 2 links r working?

6. Blog app:

```
<div>
```

A group of tags

```
</div>
```

margin: 10px 20px 30px 40px

margin: 10px 20px

margin: 10px

<https://css-tricks.com/examples/hrs/>

html:

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>DURGA Blog</title>
    <link rel="stylesheet" href="blog2.css">
    <link rel="preconnect" href="https://fonts.gstatic.com">
  <link
href="https://fonts.googleapis.com/css2?family=Orelega+One&displa
y=swap" rel="stylesheet">
  </head>
  <body>
    <div class="post">
      <div class='date'>Apr 30th 2021</div>
      <h2>This is Python Related Post</h2>
      <p class="mainp">Bacon ipsum dolor amet tongue boudin bacon
ball tip. Bresaola tenderloin shank short loin fatback tail. Doner
venison prosciutto, tri-tip strip steak ham hock ball tip tongue
tenderloin salami ham beef ribs short loin swine hamburger. Flank
bresaola venison turkey, picanha ground round tongue salami
shoulder short loin boudin cupim spare ribs leberkas pancetta. Chuck
andouille venison sausage pig porchetta doner. Flank ball tip tongue
brisket cupim shoulder pig ground round short ribs buffalo tenderloin
meatball pork chop pastrami biltong. Tenderloin pork beef, jowl
swine turkey t-bone.
      </p>
      <p>
```

Ball tip ham jerky picanha landjaeger. Ribeye ball tip shankle short ribs jerky shank strip steak cupim picanha bacon drumstick. Buffalo burgdoggen ham hock landjaeger. Pastrami turducken chicken andouille shank swine pork tenderloin, alcatra turkey frankfurter shankle hamburger rump short ribs. Buffalo chicken rump, capicola porchetta pastrami beef ribs ham hock.

</p>

<p>

Jerky kevin bacon, filet mignon ball tip picanha pancetta biltong andouille short ribs leberkas. Pastrami jerky turkey rump beef ribs venison, turducken pork loin leberkas swine spare ribs brisket. Spare ribs beef ribs fatback tongue prosciutto cupim bacon pastrami ham jerky short loin. Venison pork belly drumstick cow corned beef capicola tail jowl porchetta. Corned beef swine filet mignon short ribs beef ribs ball tip, rump salami jerky tri-tip tail burgdoggen pig ham pork chop.

</p>

<hr>

</div>

<div class="post">

<div class='date'>May 30th 2021</div>

<h2>This is Django Related Post</h2>

<p class="mainp">Bacon ipsum dolor amet tongue boudin bacon ball tip. Bresaola tenderloin shank short loin fatback tail. Doner venison prosciutto, tri-tip strip steak ham hock ball tip tongue tenderloin salami ham beef ribs short loin swine hamburger. Flank bresaola venison turkey, picanha ground round tongue salami shoulder short

loin boudin cupim spare ribs leberkas pancetta. Chuck andouille venison sausage pig porchetta doner. Flank ball tip tongue brisket cupim shoulder pig ground round short ribs buffalo tenderloin meatball pork chop pastrami biltong. Tenderloin pork beef, jowl swine turkey t-bone.

</p>

<p>

Ball tip ham jerky picanha landjaeger. Ribeye ball tip shankle short ribs jerky shank strip steak cupim picanha bacon drumstick. Buffalo burgdoggen ham hock landjaeger. Pastrami turducken chicken andouille shank swine pork tenderloin, alcatra turkey frankfurter shankle hamburger rump short ribs. Buffalo chicken rump, capicola porchetta pastrami beef ribs ham hock.

</p>

<p>

Jerky kevin bacon, filet mignon ball tip picanha pancetta biltong andouille short ribs leberkas. Pastrami jerky turkey rump beef ribs venison, turducken pork loin leberkas swine spare ribs brisket. Spare ribs beef ribs fatback tongue prosciutto cupim bacon pastrami ham jerky short loin. Venison pork belly drumstick cow corned beef capicola tail jowl porchetta. Corned beef swine filet mignon short ribs beef ribs ball tip, rump salami jerky tri-tip tail burgdoggen pig ham pork chop.

</p>

<hr>

</div>

<div class="post">

<div class='date'>June 30th 2021</div>

<h2>This is Data Science Related Post</h2>

<p class="mainp">Bacon ipsum dolor amet tongue boudin bacon ball tip. Bresaola tenderloin shank short loin fatback tail. Doner venison prosciutto, tri-tip strip steak ham hock ball tip tongue tenderloin salami ham beef ribs short loin swine hamburger. Flank bresaola venison turkey, picanha ground round tongue salami shoulder short loin boudin cupim spare ribs leberkas pancetta. Chuck andouille venison sausage pig porchetta doner. Flank ball tip tongue brisket cupim shoulder pig ground round short ribs buffalo tenderloin meatball pork chop pastrami biltong. Tenderloin pork beef, jowl swine turkey t-bone.

</p>

<p>

Ball tip ham jerky picanha landjaeger. Ribeye ball tip shankle short ribs jerky shank strip steak cupim picanha bacon drumstick. Buffalo burgdoggen ham hock landjaeger. Pastrami turducken chicken andouille shank swine pork tenderloin, alcatra turkey frankfurter shankle hamburger rump short ribs. Buffalo chicken rump, capicola porchetta pastrami beef ribs ham hock.

</p>

<p>

Jerky kevin bacon, filet mignon ball tip picanha pancetta biltong andouille short ribs leberkas. Pastrami jerky turkey rump beef ribs venison, turducken pork loin leberkas swine spare ribs brisket. Spare

ribs beef ribs fatback tongue prosciutto cupim bacon pastrami ham
jerky short loin. Venison pork belly drumstick cow corned beef
capicola tail jowl porchetta. Corned beef swine filet mignon short ribs
beef ribs ball tip, rump salami jerky tri-tip tail burgdoggen pig ham
pork chop.

</p>

<hr>

</div>

</body>

</html>

CSS:

body{

border: 20px solid #bdc3c7;

padding:20px;

width:800px;

margin:20px auto;

font-family: 'Orelega One', cursive;

}

.mainp{

border-left: 5px solid #bdc3c7;

padding-left: 5px;

}

.date{

color:#3498db;

```

    text-transform: uppercase;
    letter-spacing: 5px;
}
h2{
    color:blue;
    font-size: 2.0em;
}
.post{
    margin-bottom: 20px;
}
hr {
    height: 12px;
    border: 0;
    box-shadow: inset 0 12px 12px -12px rgba(0, 0, 0, 0.5);
}

```

HTML ---->Structure of Web Page

CSS---->To do styling for HTML elements

Bootstrap:

bootstrap.css

```

.container{
    container related styling
}

```



```
.jumbotron{  
  jumbotron related styling  
}
```

```
<form action="" method>
```

Name:

```
dljfalsjldfjas
```

```
>
```

Bootstrap is the most commonly used framework for front end development.

Bootstrap provides several predefined libraries for css and javascript.

readymade libraries we will get in bootstrap using html,css,javascript

getbootstrap.com

How to connect Bootstrap with HTML:

2 ways

1. By using CDN (Content Delivery Network)
2. Locally

bootstrap.min.css

bootstrap.css:

```
.container{  
  background: red;  
  color: white;  
}
```

bootstrap.min.css:

```
.container{ background: red; color: white;}
```

1. By using CDN (Content Delivery Network)

```
<link rel="stylesheet"  
href="https://cdn.jsdelivr.net/npm/bootstrap@4.6.0/dist/css/bootst  
rap.min.css" integrity="sha384-
```

B0vP5xmATw1+K9KRQjQERJvTumQW0nPEzvF6L/Z6nronJ3oUOFUFpCjEUQouq2+I" crossorigin="anonymous">

Note: To use CDN approach, Internet must be available.

2. Locally:

<https://getbootstrap.com/docs/4.6/getting-started/download/>

but sir if we have some specific requirements like a particular background then how can we use bootstrap???

99%-->bootstrap+ 1% you can define your own css file

- 1. Buttons**
- 2. Forms**
- 3. Navbar**
- 4. GRID**

Bhavani
durgasoftonline@gmail.com
99 2737 2737

<https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0Xulfozgly5Gs6j3j?usp=sharing>

- 1. Buttons**
- 2. Jumbotron**
- 3. Forms**
- 4. Navbar**
- 5. Grid**

1. Buttons:

buttons, anchor tags

predefined buttons provided by bootstrap.

<button type="button" class="btn btn-primary">Primary</button>

2. Jumbotron:

show case

3. Forms:

login form, registration form, enquiry form etc

form-group: grouping label and input in good format

form-control:

bootstrap will do 99% work we just need to do 1% finishing only

too good..loved bootstrap ready made template html

Developing Sample Form:

<form>

<!-- EMAIL SUBMISSION -->

<div class="form-group">

<label for="exampleInputEmail1">Email address</label>

<input type="email" class="form-control" id="exampleInputEmail1"
aria-describedby="emailHelp" placeholder="Enter email">

<small id="emailHelp" class="form-text text-muted">We'll never
share your email with anyone else.</small>

</div>

<!-- PASSWORD -->

```
<div class="form-group">
  <label for="exampleInputPassword1">Password</label>
  <input type="password" class="form-control"
id="exampleInputPassword1" placeholder="Password">
</div>
```

<!-- DROPDOWN SELECT -->

```
<div class="form-group">
  <label for="exampleSelect1">Example select</label>
  <select class="form-control" id="exampleSelect1">
    <option>1</option>
    <option>2</option>
    <option>3</option>
    <option>4</option>
    <option>5</option>
  </select>
</div>
```

<!-- MULTIPLE SELECT OPTIONS -->

```
<div class="form-group">
  <label for="exampleSelect2">Example multiple select</label>
  <select multiple class="form-control" id="exampleSelect2">
```

```
<option>1</option>
<option>2</option>
<option>3</option>
<option>4</option>
<option>5</option>
</select>
</div>
```

```
<!-- TEXT AREA -->
```

```
<div class="form-group">
  <label for="exampleTextarea">Example textarea</label>
  <textarea class="form-control" id="exampleTextarea"
rows="3"></textarea>
</div>
```

```
<!-- FILE UPLOAD INPUT -->
```

```
<div class="form-group">
  <label for="exampleInputFile">File input</label>
  <input type="file" class="form-control-file" id="exampleInputFile"
aria-describedby="fileHelp">
  <small id="fileHelp" class="form-text text-muted">This is some
placeholder block-level help text for the above input. It's a bit lighter
and easily wraps to a new line.</small>
</div>
```

<!-- RADIO BUTTONS -->

```
<fieldset class="form-group">  
  <legend>Radio buttons</legend>  
  <div class="form-check">  
    <label class="form-check-label">  
      <input type="radio" class="form-check-input"  
name="optionsRadios" id="optionsRadios1" value="option1"  
checked>  
      Option one is this and that&mdash;be sure to include why it's  
great  
    </label>  
  </div>
```

```
<div class="form-check">  
  <label class="form-check-label">  
    <input type="radio" class="form-check-input"  
name="optionsRadios" id="optionsRadios2" value="option2">  
    Option two can be something else and selecting it will deselect  
option one  
  </label>  
</div>
```



```
<div class="form-check disabled">
  <label class="form-check-label">
    <input type="radio" class="form-check-input"
name="optionsRadios" id="optionsRadios3" value="option3"
disabled>
    Option three is disabled
  </label>
</div>
```

```
</fieldset>
```

```
<!-- CHECK BUTTON -->
```

```
<div class="form-check">
  <label class="form-check-label">
    <input type="checkbox" class="form-check-input">
    Check me out
  </label>
</div>
```

```
<button type="submit" class="btn btn-primary">Submit</button>
```

```
</form>
```

- 1. Buttons**
- 2. Jumbotron**
- 3. Forms**

4. Navbar:

Navigation Bar

SORRY I WAS DEPRESSED YESTERDAY SO I DONT ATTENDED THE CLASS

SIR THINKING ABOUT JOB

SIR I AM 2017 MCA PASSED OUT

2017,2018,2019

2020--->Fresher

2017--->4 Years

3+ Years of Exp

1 or 2 calls

4 to 10 calls

70% of real time employees are fake consultancy

3 Years Ameerpet experience is more worthy than 3 Years original real time exp.

To get job several things are required, But to do the job nothing is required.

**Exception--->Fresher/Ameerpet Student
If they want to change the company**

**Infosys-->
Intermediate but not 10th class
Degree**

**Joined in the company: M.Tech
Telecom project-->Elisa
Resume
Siva**

**5+ years
telecom
some projects also**

FAQs:

Keeping FAKe

**Freshers:
-----**

1. The chance of getting job is very very low
2. The chance of getting project is very very low
3. 14th player in cricket team
4. The chance of getting onsite opportunity is very very low. almost impossible.
5. Salary hikes are very low

Experienced with God's gift:

1. The chance of getting job is very very high
2. The chance of getting project is very very high
3. 14th player in cricket team---->not there
4. The chance of getting onsite opportunity is very very high.
5. Salary hikes are very high

For any person whether experienced or fresher---->New project is always new

MCA Freshers-->BSc + 3 Years of expe

MCA --->BSc + 3 Years

M.Tech --->B.Tech+ 2 Years

4. Navbar:

Navbar--->Navigation Bar

How to create Navbar Template:

Default:

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">  
</nav>
```

Color Schemes:

```
<nav class="navbar navbar-dark bg-dark">  
  <!-- Navbar content -->  
</nav>
```

```
<nav class="navbar navbar-dark bg-primary">  
  <!-- Navbar content -->  
</nav>
```

```
<nav class="navbar navbar-light" style="background-color: red;">  
  <!-- Navbar content -->  
</nav>
```

How to add brand to the Navbar:

anchor

Heading

Image

Image and Text

Anchor:

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <a class="navbar-brand"
href="https://www.durgasoftonline.com/">DURGASOFT</a>
</nav>
```

Heading:

```
<!-- As a heading -->
<nav class="navbar navbar-light bg-light">
  <span class="navbar-brand mb-0 h1">Navbar</span>
</nav>
```

Image:

```
<a class="navbar-brand" href="https://www.durgasoftonline.com/">
  
</a>
```

<https://www.durgasoftonline.com/logo.png>

Image and Text

```
<a class="navbar-brand" href="#">
```

```
  
DURGASOFT  
</a>
```

How to add remaining items to the Navbar:

We can add remaining items to the navbar by using unordered list.
Each list item acts as Navbar item.

```
<ul class='navbar-nav mr-auto'>  
  <li class='nav-item'> <a class='nav-link' href="#">Home</a> </li>  
  <li class='nav-item'> <a class='nav-link' href="#">About Us</a>  
</li>  
  <li class='nav-item'> <a class='nav-link' href="#">Gallary</a> </li>  
  <li class='nav-item'> <a class='nav-link' href="#">Services</a> </li>  
</ul>
```

These items added by default on the left hand side.

How to add items to the Right Hand Side:

```
<ul class='navbar-nav ml-auto'>  
  <li class='nav-item'> <a class='nav-link' href="#">Contact Us</a>  
</li>  
  <li class='nav-item'> <a class='nav-link' href="#">Logout</a> </li>  
</ul>
```

How to position navbar items properly:

We have to take all navbar items inside container.

```
<div class='container'>
  navbar items
</div>
```

How to fix position of navbar:

default:

```
<nav class="navbar navbar-light bg-light">
  <a class="navbar-brand" href="#">Default</a>
</nav>
```

fixed top:

We have to add fixed-top class.

```
<nav class="navbar fixed-top navbar-light bg-light">
  <a class="navbar-brand" href="#">Fixed top</a>
</nav>
```

fixed bottom:

```
<nav class="navbar fixed-bottom navbar-light bg-light">
  <a class="navbar-brand" href="#">Fixed bottom</a>
```


</nav>

Sticky top:

```
<nav class="navbar sticky-top navbar-light bg-light">  
  <a class="navbar-brand" href="#">Sticky top</a>  
</nav>
```

How to implement Hamburger:

step-1:

We have to enclose the collapsed items inside the following div tag.

```
<div class="collapse navbar-collapse" id="navbarSupportedContent">  
  collapsed items  
</div>
```

step-2:

To add hamburger, we have to add the following <button> tag after brand.

```
<button class="navbar-toggler" type="button" data-toggle="collapse"  
  data-target="#navbarSupportedContent" aria-  
  controls="navbarSupportedContent" aria-expanded="false" aria-  
  label="Toggle navigation">  
  <span class="navbar-toggler-icon"></span>
```

</button>

Note: To work bootstrap hamburger, compulsory bootstrap javascript(js) must be required. But to work bootstrap javascript, compulsory jquery is required.

Hence we have to add js cdn and jQuery cdn in our html, but first jQuery cdn and then js cdn.

```
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"
integrity="sha384-
DfXdz2htPH0lsSSs5nCTpuj/zy4C+OGpamoFVy38MVBnE+IbbVYUew+O
rCXaRkfj" crossorigin="anonymous"></script>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@4.6.0/dist/js/bootstrap
.bundle.min.js" integrity="sha384-
Piv4xVNRyMGpqsS2by6br4gNJ7DXjqk09RmUpJ8jgGtD7zP9yug3goQfG
ll0yAns" crossorigin="anonymous"></script>
```

Bootstrap

Buttons

Jumbotron

Forms

Navbar

GRID System:

LCD

Desktop

Laptop

Tab

Mobile

To display our application layout properly on multiple devices of multiple screen sizes, Grid concept.

large screen--->more space and more number of items

small screen--->less space and less number of items

12X1 --->12 elements and each element is of 1 column length

6X2 --->6 elements and each element is of 2 column length

4X3 --->4 elements and each element is of 3 column length

$e1 \times 3 \text{ columns} + e2 \times 4 \text{ columns} + e3 \times 5 \text{ columns} = 12 \text{ columns}$

How to implement Grid:

By using 2 classes

1. row: to define row

`<div class='row'>`

2. Within the row we can define columns by using col-screensize-noOfColumns

The allowed screen sizes:

lg --->Large Size (Like Desktop screen)

md --->Medium Size (Like Laptop screen)

sm --->Small Size (Like Tab screen)

xs --->Extra Small size(Like Mobile Screen)

xl --->Extra Large (like lcd) introduced in Bootstrap 4

Extra small (xs)

Small (sm)

Medium (md)

Large (lg)

Extra large (xl)

Extra extra large (xxl)--->in Bootstrap 5

```
<div class='row'>
```

```
  <div class='col-lg-3'>Element1</div>
```

```
</div>
```

On large screen each element has to take 3 columns length.

```
<div class='row'>  
  <div class='col-lg-3'>Element1</div>  
  <div class='col-lg-3'>Element2</div>  
  <div class='col-lg-3'>Element3</div>  
  <div class='col-lg-3'>Element4</div>  
</div>
```

case-1: 3 elements

e1--->3 columns

e2--->6 columns

e3--->3 columns

```
<div class='row'>  
  <div class='col-lg-3'>Element1</div>  
  <div class='col-lg-3'>Element2</div>  
  <div class='col-lg-3'>Element3</div>  
  
</div>
```

case:

For medium or large screens each row should contain 4 elements

```
<div class='row'>
```

```
<div class='col-md-3 box'>Element1</div>
<div class='col-md-3 box'>Element2</div>
<div class='col-md-3 box'>Element3</div>
<div class='col-md-3 box'>Element4</div>
</div>
```

For all screens per row only 3 elements:

xs,sm,md,lg,xl,xxl

```
<div class='row'>
  <div class='col-4 box'>Element1</div>
  <div class='col-4 box'>Element2</div>
  <div class='col-4 box'>Element3</div>
</div>
```

xs means either xs or higher

Note: In bootstrap 4 col-xs is replaced with col-

```
<div class='col-4 box'>Element1</div>
```

How to define grid for multiple screens simultaneously:

For large screen 6 elements

For medium screen 4 elements

For small screen 3 elements

For extra small screen 2 elements

col-lg-2 col-md-3 col-sm-4 col-xs-6

```
<div class='row'>
  <div class='col-lg-2 col-md-3 col-sm-4 col-xs-6
box'>Element1</div>
  <div class='col-lg-2 col-md-3 col-sm-4 col-xs-6
box'>Element2</div>
  <div class='col-lg-2 col-md-3 col-sm-4 col-xs-6
box'>Element3</div>
  <div class='col-lg-2 col-md-3 col-sm-4 col-xs-6
box'>Element4</div>
  <div class='col-lg-2 col-md-3 col-sm-4 col-xs-6
box'>Element5</div>
  <div class='col-lg-2 col-md-3 col-sm-4 col-xs-6
box'>Element6</div>
</div>
```

<https://drive.google.com/drive/folders/1GoWlJyZ3P2AWjca0Xulfozgly5Gs6j3j?usp=sharing>

Buttons

Jumbotron

Forms

Navbar and Hamburger

GRID

Nested GRID:

Grid inside Grid

```
<div class="container">
  <div class='row'>
    <div class='col-6 box'>
      <div class="row ">
        <div class='col-3 box'>Element1A</div>
        <div class='col-3 box'>Element1B</div>
        <div class='col-3 box'>Element1C</div>
        <div class='col-3 box'>Element1D</div>
      </div>
    </div>
    <div class='col-6 box'>
      <div class="row ">
        <div class='col-4 box'>Element2A</div>
        <div class='col-4 box'>Element2B</div>
        <div class='col-4 box'>Element2C</div>
      </div>
    </div>
  </div>
```

Bootstrap Photo Gallery:

```
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@4.6.0/dist/css/bootst
rap.min.css" integrity="sha384-
B0vP5xmATw1+K9KRQjQERJvTumQW0nPEzvF6L/Z6nronJ3oUOFUFpCj
EUQouq2+I" crossorigin="anonymous">
```

```
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"
integrity="sha384-
DfXdz2htPH0lsSSs5nCTpuj/zy4C+OGpamoFVy38MVBnE+IbbVYUew+O
rCXaRkfj" crossorigin="anonymous"></script>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@4.6.0/dist/js/bootstrap
.bundle.min.js" integrity="sha384-
Piv4xVNRyMGpqsS2by6br4gNJ7DXjqk09RmUpJ8jgGtD7zP9yug3goQfG
ll0yAns" crossorigin="anonymous"></script>
```

large screen-->4 images
remaining cases --->2 images

sir class=" col-lg-3 col-6 " explain this sir once again ?

DURGA BAR Application by using Bootstrap - Navbar, Jumbotron, Grid
and font awesome icons

HTML

CSS

Bootstrap

Durga Dating App

```
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"
integrity="sha384-
DfXdz2htPH0lsSSs5nCTpuj/zy4C+OGpamoFVy38MVBnE+IbbVYUew+O
rCXaRkfj" crossorigin="anonymous"></script>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@4.6.0/dist/js/bootstrap
.bundle.min.js" integrity="sha384-
Piv4xVNRyMGpqsS2by6br4gNJ7DXjqk09RmUpJ8jgGtD7zP9yug3goQfG
ll0yAns" crossorigin="anonymous"></script>
```

unsplash

<p>Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do
eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim
ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut

aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.</p>

<h2>We are very specialized for</h2>

Late Marriages

Early Marriages

xyz Marriages

abc Marriages

zzz Marriages

<https://www.w3schools.com/howto/>

i want to ask that we have already java python and other languages .then what is the requirement of the scripting laguages??

HTML

CSS

Bootstrap

Java Script:

To add functionality to the HTML components

Full Stack---->Any database+Front+Java | Django

MEAN Stack---->MongoDB+Express JS+Angular+Node

MERN Stack----->MongoDB+Express JS+React+Node

Full stack web development with Python and Django

- 1. Java Script Developer's console**
- 2. 5 Basic java script primitive data types**
- 3. Declare variables**

- 4. The 3 most commonly used js functions**

1. Java Script Developer's console:

10+20

snippets-->Java Script Developer's console

Python Shell

Python IDLE

How to launch:

Browser-->Right click-->Inspect--->Console

ctrl+shift+j or F12

To clear console: clear() or ctrl+l

2. 5 Basic java script primitive data types

1. number

10

10.5

-10

10+20

10-20

20/10

20*10

102**

2. string:

any sequence of characters either within single quotes or double quotes

'durga'

"durga"

+ operator==>concatenation

'durga'+'soft'--->durgasoft

'durga'+10--->durga10

'durga'+true--->durgatrue

If both arguments are number type then + operator acts as arithmetic addition operator.

If atleast one argument is string type, then concatenation operator

We can use escape characters also

'durga\nsoft'

'durga\tsoft'

How to find length of the string:

s.length

How to access character of a string:

index

s[2]
s[10]

boolean:

true/false
True/False

number
string
boolean

x = 10

var name = 'durga'
var age = 60
var isMarried = true

Dynamically Typed Language

variable naming convention in javascript:

studentMobileNumber--->camel case(java | Javascript)

student_mobile_number-->snake case(Python)

student-mobile-number--->Kebab Case(LISP)

- 1. number**
- 2. string**
- 3. boolean**
- 4. undefined**

Variables are declared but not initialized or if the value is not available -->undefined

5. null

null means nothing

If the value of a variable is null means it is not pointing to any value.

var currentplayer = 'durga'

game completed

currentplayer = null

<https://flowup.000webhostapp.com/> --->Ram

<https://manishp.000webhostapp.com/> --->Manish

<https://jalapic-orifices.000webhostapp.com/admins/login>

The 3 most commonly used methods of java script:

- 1. alert()**
- 2. console.log()**
- 3. prompt()**

1. alert():

To display alerts to the end user.

alert('Hello Good Evening')
alert(100)
alert(true)

alert('Friends,,,prioritize your activities... short cut for success:'+100)

alert(100+250)

sir alert('10'+ '20') here concat or arithmetic

2. console.log()

To print messages to the developers console

console.log('Hello Friends...')

```
console.log(10*20)
```

console messages not meant for end user.

3. prompt()

```
name = input('Enter Your Name:')--->Python
```

To get input from the end user...

```
var name = prompt('Enter Your Name:')
```

```
var quote = prompt('Enter Your Favourite Quotation:')
```

```
alert('Hello'+name+', Thanks for your caption:\n'+quote)
```

1. alert()

2. console.log()

3. prompt()

separate file: .js file

How to write javascript code to a separate file and connecting to HTML:

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@4.6.0/dist/js/bootstrap
.bundle.min.js" integrity="sha384-
Piv4xVNRyMGpqrS2by6br4gNJ7DXjqk09RmUpJ8jgGtD7zP9yug3goQfG
ll0yAns" crossorigin="anonymous"></script>

<link type href='demo.css'>

<link rel="stylesheet" href="dating1.css">

<script type='text/javascript' src='test.js'></script>

Inside head--->first js code and then html body

inside body--->it will be executed with body

<https://annaamma.000webhostapp.com/>

<https://lawless-screens.000webhostapp.com/>

today i have completed app to my client in upwork sir and completed and got psymeny sir

<http://techbetainfo.rf.gd/sayikumar.html> used lightbox for image view sir

Sir please look at the webpages I made at <http://www.hazeinfo.net>

Operators:

1. Arithmetic Operators:

+, -, *, /, %, **

//--->acts as a comment but not floor division operator

2. Comparison operators:

<, <=, >, >=, ==, !=, ===

10<20

10<=20

10>20

10>=20

10==20

10 !=20

Difference between == and ===:

== operator is equality operator

=== operator is strict equality operator

10 == 10

10 == '10'

10 == 10.0

Here only values are important but not type.

Both values and type are important

10 === 10

10 === '10'

10 === 10.0

== operator is equality operator where values are important but not types

=== operator is strict equality operator where both types and values are important

=== recommended to use.

true == '1'

false == '0'

null == undefined

true === '1'

false === '0'

null === undefined

NaN(Not a Number)

0/0 --->NaN

10<NaN

10<=NaN

10>NaN

10>=NaN

10 == NaN

NaN == NaN

In all these cases we will get false

10 != NaN

NaN != NaN

true answer

0/1 == NaN

Logical Operators:

&& --->AND

|| -->OR

! ---->NOT

x && y --->if both are true then only it returns true.

x || y --->if atleast one is true then it returns true

For logical operators:

zero --->>false

non-zero --->>true

empty string --->>false

non-empty string --->>true

null,undefined,NaN --->>false

x = 10

y = 20

x<10 && x!= 5 --->>false

y>9 || x==10 ---->>true

!(x == y) ---->>true

!(x == '10' || x === y)&& !(y != 8 && x<= y) --->>false

Conditional statements:

1. if

2. if -else

3. if-else if

Syntax:

```
if (b){  
  action if b is true  
}  
else{  
  action if b is false  
}
```

```
x = prompt('Enter Any number to check:')  
if (x == 10)
```

```
number('10')+'20'
```

```
'The given Number is:'+x+' It is Odd'--->string
```

```
console.log(string)  
console.log('The given Number is:'+x+' It is even')
```

```
'The given Number is:'+10+' It is even'
```

```
'The given Number is:10+' It is even'
```

```
'The given Number is:10 It is even'
```


Matrimonial website:

age from the end user

18 to 60 years

```
var age = Number(prompt('Enter Your age:'))
if (age >60 ){
    alert('Please wait some more time, definitely you will get best
match...')
}
else if(age<18){
    alert('Your age already crossed marriage age, no chance of getting
marriage')
}
else{
    alert('Thanks for registration...You will get match details soon by
email')
}
```

read brand from the end user and provide meaningful message.

```
var brand = prompt('Enter Your Favourite Brand:')
if (brand == 'KF'){
    alert('It is childrens brand')
}
else if(brand == 'KO'){
```

```
    alert('It is too light')
}
else if(brand == 'RC'){
    alert('It is not that much kick')
}
else if(brand == 'FO'){
    alert('Buy one get One FREE')
}
else{
    alert('Other brands are not recommended')
}
```

Q. Number Guess application

read a number form the end user

var sno=4

var num = Number(prompt('Enter Your Guess between 1 to 9:'))

if (num > sno){

alert('It is too high.. Guess Again')

}

else if (num < sno){

alert('It is too Low...Guess Again')

}

else{

alert('Congratulation...Your Guess is right')

}

Iterative Statements:

to execute a group of statements iteratively... iterative statements

DRY Principle: Don't Repeat Yourself

2 iterative statements

1. while loop

2. for loop

1. while loop:

as long as some condition is true, then we should go for while loop.

Syntax:

```
while(condition){  
    body  
}
```

eg-1: To print hello 10 times to the console

eg-2: To print first 10 numbers

eg-3: To print each character present in the given string?

durga

d

u

r

g

a

```
var s = prompt('Enter Any String:')
var i = 0
while (i < s.length){
  console.log('The Character Present at index:' + i + ' is:' + s[i])
  i++
}
```

eg: To print nth table?

2 X 1 = 2

2 X 2 = 4

```
var n = Number(prompt('Enter n value:'))
var i = 1
while (i <= 10){
  console.log( n + ' X ' + i + ' = ' + (n*i))
  i++
}
```

Q. To print all numbers divisible by 3 And 5 between 5 and 100.

var n = 5

```
while (n <= 100){  
  if( n%3 == 0 && n%5 == 0){  
    console.log(n)  
  }  
  n++  
}
```

If we don't know number of iterations in advance: while loop

Iterative Statements: for loop

If we know number of iterations in advance then we should go use for loop.

Syntax:

```
for(initialization_section;conditional_check;increment_decrement_section){  
  body  
}
```

eg-1: To print 'Hello' 10 times.

```
for(var i = 0;i<10; i++){
```

```
console.log('Hello')
}
```

eg-2: To print numbers from 1 to 10???

```
for(var i = 1;i<=10; i++){
  console.log(i)
}
```

eg-3: To print all numbers which are divisible by 7 from 1 to 100?

```
for(var i = 1;i<=100; i++){
  if(i%7 == 0){
    console.log(i)
  }
}
```

eg-4: To print each character from the given string?

'durga'

0,1,2,3,4 length-->5

```
var str = prompt('Enter Any String:')
for(var i =0; i< str.length; i++){
  console.log(str[i])
}
```

while vs for:

If we dont know number of iterations in advance and as long as some condition is true, execute body...while loop

If we know number of iterations in advance--->for loop

Secret Agent Application:

Rules:

1. The first character of the name should be 'd'
2. The last character of favourite actor should be 'r'
3. The lucky number should be 7.
4. The length of favourite dish(no of characters) should be ≥ 6

'7' == 7

ameer

5

0,1,2,3,4

actor[4]

```
var name = prompt('Enter Your Name:')
var actor = prompt('Enter Your Favourite Actor Name:')
var lucky = prompt('Enter Your Lucky Number:')
var dish = prompt('Enter Your Favourite Dish:')

var nameCondition = false
var actorCondition = false
var luckyCondition = false
var dishCondition = false

if (name[0] == 'd'){
    nameCondition = true
}

if (actor[actor.length -1] == 'r'){
    actorCondition = true
}

if (lucky == 7){
    luckyCondition = true
}

if (dish.length >= 6){
    dishCondition = true
}

alert('Hello '+name+'\nThanks for your information!!!')
if( nameCondition && actorCondition && luckyCondition &&
dishCondition ){
```



```
console.log('Hello Secret Agent, Our next operation is:')  
console.log('We have to kill atleast 10 sleeping students in the class  
room because these people are burdent to the country')  
}
```

sir actor.length-1 one time tell

```
actor = 'ameer'  
actor.length --->5  
actor[0] --->a  
actor[1] --->m  
actor[2] --->e  
actor[3] --->e  
actor[4] --->r
```

actor[actor.length-1] ---->last character of actor

actor[0]

Functions:

```
function functionName(arguemnts){  
    line-1  
    line-2  
    line-3
```

```
    line-4
    line-5
}
```

```
functionName()
functionName()
functionName()
functionName()
```

Code reusability

Syntax:

```
function functionName(arguemnts){
    body
    return value
}
```

eg-1: To print Good evening

Function with arguments:

inputs to the function.

eg: to wish person by name

```
function wish(name){
    console.log('Hello '+name+'... Good Evening!!!!')
```

```
}
```

```
wish('Durga')
```

```
wish('Veeru')
```

Function with default arguments:

```
function wish(name='Guest'){  
  console.log('Hello '+name+'... Good Evening!!!')  
}
```

```
wish()
```

```
wish('Veeru')
```

Function with return values:

1. To find square of given number???

```
function squareIt(num){  
  return num*num  
}
```

```
var result = squareIt(10)
```

```
console.log('The Square of 10 is:'+result)
```

```
console.log('The Square of 20 is:'+squareIt(20))
```

2. to take 2 numbers as arguments and return sum???

```
function sum(num1,num2){  
  return num1+num2  
}
```

```
console.log('The sum of 10 and 20:'+sum(10,20))  
console.log('The sum of 100 and 200:'+sum(100,200))
```

**3. WAJF to take a string as argument and return capitalized string?
durga ==>Durga**

```
function capitalize(str){  
  result = str[0].toUpperCase()+str.slice(1)  
  return result  
}
```

```
console.log(capitalize('apple'))
```

4. WAJF to check whether the given number is even or not????

```
function isEven(num){  
  if (num%2 == 0){  
    return true  
  }  
  else{  
    return false  
  }  
}
```

```
}  
}
```

```
console.log(isEven(15))  
console.log(isEven(10))
```

WAJF to find factorial of given number?

```
function factorial(num){  
  result = 1  
  for(var i = 2; i<= num; i++){  
    result = result*i  
  }  
  return result  
}
```

```
console.log('The factorial of 4 is:'+factorial(4))  
console.log('The factorial of 5 is:'+factorial(5))
```

WAJF to convert from snake case to kebab case?

```
durga_soft_solutions  
durga-soft-solutions
```

```
function snakeToKebab(str){  
  var newStr = str.replaceAll('_', '-')  
  return newStr
```

```
}
```

```
console.log(snakeToKebab('durga_software_solutions'))
```

Java Script Scopes:

1. Global scope

2. Local scope

The variables which are declared outside of function are having global scope and available for all functions.

The variables which are declared inside function are having local scope and available only for that function.

```
var x = 777 //Global variable
```

```
function f1(){
```

```
  x=10 // It is not new declaration and it refers global copy
```

```
  console.log(x)
```

```
}
```

```
function f2(){
```

```
  console.log(x)
```

```
}
```

f1()

f2()

Q. If local and global variables having same name then within the function local variable will get high priority. How to access global variable value??

x = 777 //Global variable

function f1(){

var x=10

console.log(x) //10

console.log(???) //777

}

f1()

Higher Order functions:

The functions which can take another function as argument, such type of special functions are called higher order functions.

setInterval(function, time_in_milliseconds)

clearInterval(1)

Anonymous functions:

nameless function

just for instant use

```
setInterval(function(){console.log('Anonymous function');}, 4000)
```

```
x = 777
function f1()
{
  var x = 10
  console.log(x)
  console.log(this.x)
}
f1()
```

```
x = 777
function f1()
{
  var x = 10
  console.log(x)
  console.log(this.x)
  console.log(window.x)
}
f1()
```


How to access global variables inside a function, if both global and local variables have same name?

```
x = 777
function f1()
{
  var x = 10
  console.log(x)
  console.log(this.x)
  console.log(window.x)
}
f1()
```

Coding Examples :

Problem-1: sleep_in

Write a function called sleep_in that takes 2 boolean parameters: weekday and vacation.

The parameter weekday is True if it is a weekday, and the parameter vacation is True if we are on vacation. We sleep in if it is not a weekday or we're on vacation. Return True if we sleep in.

```
sleep_in(false, false) --> true
sleep_in(true, false) --> false
sleep_in(false, true) --> true
sleep_in(true, true) --> true
```

```

function sleep_in(weekday,vacation){
  if (weekday == false) {
    return true
  }
  else if(vacation == true){
    return true
  }
  return false
}
console.log('Is Employee Sleeping:'+ sleep_in(true,true)) //true
console.log('Is Employee Sleeping:'+ sleep_in(true,false))//false
console.log('Is Employee Sleeping:'+ sleep_in(false,true))//true
console.log('Is Employee Sleeping:'+ sleep_in(false,false))//true

```

2nd way:

```

function sleep_in(weekday,vacation){
  return !weekday || vacation
}
console.log('Is Employee Sleeping:'+ sleep_in(true,true)) //true
console.log('Is Employee Sleeping:'+ sleep_in(true,false))//false
console.log('Is Employee Sleeping:'+ sleep_in(false,true))//true
console.log('Is Employee Sleeping:'+ sleep_in(false,false))//true

```

Problem-2: monkey_trouble:

We have two monkeys, a and b, and the parameters a_smile and b_smile indicate if each is smiling. We are in trouble if they are both smiling or if neither of them is smiling. Return True if we are in trouble.

```
monkey_trouble(true, true) --> true
monkey_trouble(false, false) --> true
monkey_trouble(true, false) --> false
```

```
function monkey_trouble(a_smile, b_smile){
  if ((a_smile == true && b_smile == true) || (a_smile == false &&
b_smile == false)){
    return true
  }
  else{
    return false
  }
}
console.log('Is Person in trouble:'+monkey_trouble(true,true)); //true
console.log('Is Person in trouble:'+monkey_trouble(false,false));
//true
console.log('Is Person in trouble:'+monkey_trouble(true,false));//false
```

2nd way:

```
function monkey_trouble(a_smile, b_smile){
  return (a_smile && b_smile) || (!a_smile && !b_smile)
}
console.log('Is Person in trouble:'+monkey_trouble(true,true)); //true
console.log('Is Person in trouble:'+monkey_trouble(false,false));
//true
console.log('Is Person in trouble:'+monkey_trouble(true,false));//false
```

Problem-3: string_times :

Given a string and a non-negative int n, return a larger string that is n copies of the original string. ie string repeatition operator.

```
string_times('Hi', 2) --> 'HiHi'
string_times('Hi', 3) --> 'HiHiHi'
string_times('Hi', 1) --> 'Hi'
```

```
function string_times(str,n){
  result=""
  count=1
  while (count <= n){
    result = result+str
    count++
  }
}
```

```
    return result
}
console.log(string_times('durga',3));
console.log(string_times('hello',2));
console.log(string_times('hi',1));
```

```
function string_times(str,n){
    result=""
    for (var i = 0; i < n; i++) {
        result += str
    }
    return result
}
console.log(string_times('durga',3));
console.log(string_times('hello',2));
console.log(string_times('hi',4));
```

Problem-4: lucky_sum :

Given 3 int values, a b c, return their sum. However, if one of the values is 13 then it does not count towards the sum and values to its right do not count. So for example, if b is 13, then both b and c do not count.

lucky_sum(1, 2, 3) --> 6

lucky_sum(1, 2, 13) --> 3
lucky_sum(1, 13, 3) --> 1
lucky_sum(13, 13, 3) --> 0

```
function lucky_sum(a,b,c){  
  if (a==13){  
    return 0;  
  }  
  else if(b==13){  
    return a;  
  }  
  else if(c == 13){  
    return a+b  
  }  
  else{  
    return a+b+c  
  }  
}  
console.log(lucky_sum(1, 2, 3)); //6  
console.log(lucky_sum(1, 2, 13)); //3  
console.log(lucky_sum(1, 13, 3)); //1  
console.log(lucky_sum(13, 13, 3)); //0
```

ternary operator:

result = (condition)? x: y

if condition is true then x value will be considered else y value will be considered.

2nd way:

```
function lucky_sum(a,b,c){  
  return (a==13)? 0:(b==13)?a:(c==13)?a+b:a+b+c  
}  
console.log(lucky_sum(1, 2, 3)); //6  
console.log(lucky_sum(1, 2, 13)); //3  
console.log(lucky_sum(1, 13, 3)); //1  
console.log(lucky_sum(13, 13, 3)); //0  
console.log('Superb way!!!');
```

Problem-5: caught_speeding

You are driving a little too fast, and a police officer stops you. Write code to compute the result, encoded as an int value: 0=no ticket, 1=small ticket, 2=big ticket. If speed is 60 or less, the result is 0. If speed is between 61 and 80 inclusive, the result is 1. If speed is 81 or more, the result is 2. Unless it is your birthday -- on that day, your speed can be 5 higher in all cases.

caught_speeding(60, False) --> 0

caught_speeding(65, False) --> 1
caught_speeding(65, True) --> 0

```
function caught_speeding(speed, isBirthday){  
  if (isBirthday){  
    speed = speed-5  
  }  
  if (speed <= 60){  
    return 0  
  }  
  else if (speed >= 61 && speed <= 80) {  
    return 1  
  }  
  else{  
    return 2  
  }  
}  
console.log('Getting Ticket with number:'+caught_speeding(60,false));  
//0  
console.log('Getting Ticket with  
number:'+caught_speeding(65,false));//1  
console.log('Getting Ticket with  
number:'+caught_speeding(65,true));//0
```

60--->no ticket

65--->no ticket if birthday

JavaScript Arrays:

An indexed collection of elements.

x=10

y=20

z=30

numbers = [10,20,30]

Accessing elements of array by using index:

var friends = ['sunny','bunny','chinny','vinny','pinny']

var friends = ['sunny','bunny','chinny','vinny','pinny']

console.log(friends[0]);//sunny

console.log(friends[2]);//chinny

console.log(friends[30]);

if we are trying to access array element with out of range index, then we won't get any error and just we will get undefined.

negative indexing is not supported.

update array elements by using index:

var friends = ['sunny','bunny','chinny','vinny','pinny']

```
friends[0] = 'durga'
```

How to add new elements to the array by using index:

```
var friends = ['sunny','bunny','chinny','vinny','pinny']
```

```
var friends = ['sunny','bunny','chinny','vinny','pinny']
```

```
friends[5]='ravi'
```

```
friends[40]='shiva'
```

```
console.log(friends);
```

How to create an empty array:

```
var numbers = []
```

```
var numbers = new Array()
```

How to find length of array:

length variable

```
numbers.length
```

Note: javascript array can contain heterogeneous elements also.

```
var numbers = [10,'durga',10.5,null]
```

```
console.log(numbers);
```

Important methods related to javascript arrays:

1. `push()`
2. `pop()`
3. `unshift()`
4. `shift()`
5. `indexOf()`
6. `slice()`

1. `push()`

We can use `push()` method to add elements at the end of array. After adding this element it returns the length of the array.

```
var numbers = [10,20,30,40,50]
console.log(numbers.push(9999))
console.log(numbers);
```

2. `pop()`:

To remove and return last element of the array.

```
var numbers = [10,20,30,40,50]
console.log(numbers.pop())
console.log(numbers.pop())
console.log(numbers.pop())
console.log(numbers);
```

3. unshift():

To add element at beginning of the array. counter part of push operation.

```
var numbers = [10,20,30,40,50]
numbers.unshift(9999)
console.log(numbers);
```

4. shift:

counter part to pop operation.

To remove and return first element from the array.

```
var numbers = [10,20,30,40,50]
console.log(numbers.shift());
console.log(numbers);
```

5. indexOf():

We can use indexOf() to find index of specified element.

if the element present multiple times then it will return index of first occurrence.

If the specified element is not available then -1

```
var numbers = [10,20,30,40,50,30]
console.log(numbers.indexOf(70));
```

6. slice():

To get part of the array

slice(begin,end)---->returns array elements from begin index to end-1 index.

slice()---->returns all elements.

```
var numbers = [10,20,30,40,50,60,70,80]
num1 = numbers.slice(1,5);
console.log(num1)
num2 = numbers.slice()
console.log(num2);
```

1. push() --->To add element at end of the array
2. pop() --->To remove and returns last element
3. unshift() --->To add element at beginning of the array
4. shift() --->To remove and return first element
5. indexOf()--->To return index of first occurrence of the given element
6. slice()--->to get part of the array from begin index to end-1 index.

Multi dimensional arrays/Nested arrays

```
-----  
var numbers = [[10,20,30],[40,50,60],[70,80,90]]  
console.log(numbers);  
console.log(numbers[0][2]);  
console.log(numbers[2][1]);
```

Book Management Application:

```
-----  
books = []  
input = prompt('Which operation you want to  
perform[add | list | exit]:')  
while (input != 'exit'){  
  if (input == 'add'){  
    newBook = prompt('Enter Name of the book to add:')  
    books.push(newBook)  
  }  
  else if(input == 'list'){  
    console.log('The List of available books:');  
    console.log(books);  
  }  
  else{  
    console.log('Invalid Option');  
  }  
  input = prompt('Which operation you want to  
perform[add | list | exit]:')  
}  
console.log('Thanks for using our application');
```

How to retrieve elements of the array:

We can retrieve elements of the array by using the following ways:

- 1. while loop**
- 2. for loop**
- 3. for-of loop**
- 4. forEach loop**

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How to retrieve elements of the array:

We can retrieve elements of the array by using the following ways:

- 1. while loop**
- 2. for loop**
- 3. for-of loop**
- 4. forEach method**

1. while loop:

```
nums = [10,20,30,40,50]  
var i =0  
while(i<nums.length){
```

```
    console.log(nums[i]);  
    i++; // i = i+1  
}
```

2. for loop:

```
nums = [10,20,30,40,50,60]  
for (var i = 0; i < nums.length; i++) {  
    console.log(nums[i])  
}
```

3. for-of loop:

```
nums = [10,20,30,40,50,60,70]  
for (num of nums) {  
    console.log(num)  
}
```

4. forEach() method:

It is specially designed method to retrieve elements of array.

arrayObject.forEach(function)

```
heroines = ['sunny','mallika','katrina','kareena','kajal']  
function printElement(element) {
```



```
console.log('*****');
console.log(element);
console.log('*****');
}
heroines.forEach(printElement)
// for each element present in the heroines array, printElement
function will be executed by passing that element as argument.
```

eg-2:

```
heroines = ['sunny','mallika','katrina','kareena','kajal']
```

```
heroines.forEach(function(element) {
  console.log('*****');
  console.log(element);
  console.log('*****');
})
// for each element present in the heroines array, printElement
function will be executed by passing that element as argument.
```

```
heroines = ['sunny','mallika','katrina','kareena','kajal']
heroines.forEach(console.log)
// for each element present in the heroines array, console.log()
function will be executed by passing that element as argument.
```

```
heroines = ['sunny','mallika','katrina','kareena','kajal']
```

```
heroines.forEach(alert)
```

// for each element present in the heroines array, alert() function will be executed by passing that element as argument.

for loop vs forEach():

1. for loop is general purpose loop and applicable everywhere. But forEach() function is applicable only for arrays.

2. By using for loop, we can move either in the forward direction or in the backward direction. But by using forEach() method we can move only in forward direction.

```
heroines = ['sunny','mallika','katrina','kareena','kajal']
```

```
for (var i = 0; i < heroines.length; i++) {
```

```
  console.log(heroines[i])
```

```
}
```

```
console.log('*****');
```

```
for (var i = heroines.length-1; i >= 0; i--) {
```

```
  console.log(heroines[i])
```

```
}
```

How to delete elements based on index:

splice() function

arrayobject.splice(index,number of elements)

arrayobject.splice(2,1)

arrayobject.splice(2,2)

heroines = ['sunny','mallika','katrina','kareena','kajal']

heroines.splice(2,2)

console.log(heroines);

heroines[2]='xyz'

How to insert elements at specified index:

heroines = ['sunny','mallika','katrina','kareena','kajal']

heroines.splice(2,0,'xyz','aaa')

console.log(heroines.indexOf('xyz'));//2

console.log(heroines.indexOf('aaa'));//3

Immutability vs mutability:

arrays are mutable. vs but string objects are immutable.

Once we creates an array, we can perform any changes in that object.

Once we create a string object, we are not allowed to change its content in the existing object. If we are trying to change, with those changes a new object will be created.

```
heroines = ['sunny','mallika','katrina','kareena','kajal']  
heroines[0]='xyz'  
console.log(heroines);
```

```
name = 'sunny'  
name[0]='B'  
console.log(name);
```

```
name = 'sunny'  
name.toUpperCase()  
console.log(name);
```

Mutability means changeable whereas immutability means non-changeable.

Q1. Write a javascript function to take an array as argument and print its elements in reverse order?

```
function reverse(array) {
```

```

for (var i = array.length-1; i >=0; i--) {
    console.log(array[i]);
}
reverse([10,20,30,40])
reverse(['A','B','C','D','E'])

```

Q2. Write a javascript function to take an array as argument and checks whether elements are identical or not?

```

function identical(array) {
    first=array[0]
    for (var i = 1; i < array.length; i++) {
        if (array[i] != first) {
            return false
        }
    }
    return true
}
console.log(identical([10,10,10,10]))
console.log(identical([10,10,10,30,10,10]))

```

Q3. Write a javascript function to take an array as argument and finds max value?

```

function findMax(array) {

```

```
max=array[0]
for (var i = 1; i < array.length; i++) {
  if (array[i] > max) {
    max = array[i]
  }
}
return max
}
console.log(findMax([10,5,4,15,9]))
```

To find min and max present in the given number?

```
function findMinMax(num) {
  max=num.charAt(0)
  min=num.charAt(0)
  for (var i = 1; i < num.length; i++) {
    if(num.charAt(i) < min){
      min=num.charAt(i)
    }
    if(num.charAt(i) > max){
      max=num.charAt(i)
    }
  }
  alert('The Min is:'+min+'\nThe Max is:'+max)
}
num = prompt('Enter Any Number:')
findMinMax(num)
```

Book Management Application:

```
books = []
input = prompt('Which Operation You want to
perform[add | delete | list | exit]')
while (input != 'exit'){
  if (input == 'add'){
    addBook();
  }
  else if( input == 'list' ){
    listBooks();
  }
  else if(input == 'delete'){
    deleteBook();
  }
  else{
    console.log('Invalid Option, Please Enter Valid Option');
  }
  input = prompt('Which Operation You want to
perform[add | delete | list | exit]')
}
```

```
console.log('Thanks for using our application');
```

```
function addBook(){  
    newBook = prompt('Enter the name of the book to add')  
    books.push(newBook)  
    console.log('Specified Book added');  
}  
function listBooks(){  
    console.log('List of Available Books:');  
    for (book of books) {  
        console.log(book);  
    }  
}  
function deleteBook(){  
    name = prompt('Enter the name of book to delete:');  
    index = books.indexOf(name)  
    if (index == -1) {  
        console.log('Specified Book is not available');  
    }  
    else{  
        books.splice(index,1)  
        console.log('Specified Book deleted');  
    }  
}
```

JavaScript Objects:

xml

json/yaml

```
import requests
r=requests.get('https://api.coindesk.com/v1/bpi/currentprice.json')
data=r.json()
print('Hello Friends, If you required bit coin price,please contact me')
print(f'As on {data['time']['updated']}, Bitcoin price:
${data['bpi']['USD']['rate']}")
```

JavaScript Objects:

Array--->we can store a group of individual objects.

Object--> A group of key-value pairs

Syntax:

```
var variableName = {
    key1:value1,
    key2:value2,
    key3:value3,
    ...
}
```

```
var movie={
    name:'Bahubali',
```

```
year:2016,  
hero:'Prabhas',  
latest_movie  
latestMovie  
}  
console.log(typeof(movie))  
console.log(movie)
```

How to access values from Object:

1. obj["key"]
eg: movie['name']

2. obj.key
eg: movie.name

How to create and initialize javascript objects?

```
var nums={}  
var nums = new Object()
```

To add key-values to this object:

1st way:
nums['fno']=100
nums['sno']=200

```
nums['tno']=300
```

2nd way:

```
nums.fno=400
```

If the specified key is already available, then old value will be replaced with new value.

How to update values:

```
nums['fno']=700
```

```
nums.fno=800
```

Iterating Objects:

```
var nums = {fno:100,sno:200,tno:300}
for(key in nums){
    //console.log(key) //only keys
    //console.log(nums[key]); only values
    console.log(key+'----->'+nums[key]);
}
```

Nested Objects and Arrays:

Inside objects--->arrays

Inside arrays --->objects

```
var movies = [  
  {name:'Bahubali',year:2016,hero:'Prabhas'},  
  {name:'Spider',year:2017,hero:'MaheshBabu'},  
  {name:'Sanju',year:2018,hero:'Ranveer'}  
]
```

`movies[0].name`

`movies[3].hero`

`movies.name`

which name displayed sir

```
var numbers = {  
  fg: [10,20,30],  
  sg: [40,50,60],  
  tg: [70,80,90]  
}
```

`numbers.sg[1]`

Object Methods:

Inside object, we can take methods also.

```
var myObj={  
  A:'Apple',  
  B:'Banana',  
  m1:function(){  
    console.log('Object Method')  
  }  
}
```

myObj.m1()

this keyword:

this means current object

```
var myObj={  
  A:'Apple',  
  B:'Banana',  
  m1:function(){  
    console.log('First Fruit:'+this.A)  
    console.log('Second Fruit:'+this.B)  
  }  
}
```

It is possible to refer already existing function as object method.

```
function demo(){  
  console.log('Demo Function');  
}
```

```
var movie={  
  name:'Bahubali',  
  year:2016,  
  hero:'Prabhas',  
  getInfo:demo  
}
```

movie.getInfo() //demo() function will be executed.

Inside demo we can access the properties of object.

```
function demo(){  
  console.log('Movie Name:'+this.name);  
  console.log('Year of Release:'+this.year);  
  console.log('Hero Name:'+this.hero);  
}
```

```
var movie={
```

```
name:'Bahubali',  
year:2016,  
hero:'Prabhas',  
getInfo:demo  
}
```

```
movie.getInfo()
```

```
demo()
```

```
var movie={  
  name:'Bahubali',  
  year:2016,  
  hero:'Prabhas',  
  getInfo: function demo(){  
    console.log('Movie Name:'+this.name);  
    console.log('Year of Release:'+this.year);  
    console.log('Hero Name:'+this.hero);  
  }  
}
```

```
var movie={  
  name:'Bahubali2',  
  year:2018,  
  hero:'Prabhas',  
  getInfo(){  
    console.log('Movie Name:'+this.name);  
    console.log('Year of Release:'+this.year);  
    console.log('Hero Name:'+this.hero);  
  }  
}
```

```
var movie={  
  name:'Bahubali2',  
  year:2018,  
  hero:'Prabhas',  
  getInfo(a){  
    console.log('Movie Name:'+this.name);  
    console.log('Year of Release:'+this.year);  
    console.log('Hero Name:'+this.hero);  
    console.log('Rating:'+a);  
  }  
}
```



```
movie.getInfo(5)
```

I watched Bahubali and Movie is Hit!!!

I have not seen Sanju and Movie is Hit!!!

I watched Spider and Movie is Flop!!!

```
result='I '
```

```
var fruits=['Apple','Banana','Orange']
fruits.forEach(function(fruit){
  if (fruit=='Apple')
  {
    console.log('It is too good for health')
  }
})
```

Mini Application:

```
var movies =[
  {name:'Bahubali',isWatched:true,isHit:false},
  {name:'Sanju',isWatched:true,isHit:false},
  {name:'Spider',isWatched:true,isHit:false},
  {name:'VakeelSab',isWatched:true,isHit:true},
]
```

```
movies.forEach(function(movie){
  result ='I '
  if (movie.isWatched) {
    result=result+'watched '
  }
  else{
    result=result+'have not seen '
  }
  result=result+movie.name+' and Movie is '
  if (movie.isHit) {
    result=result+'Hit!!!'
  }
  else{
    result=result+'Flop!!!'
  }
  console.log(result);
})
```

HTML

CSS

Bootstrap

JS

DOM

DOM--->Document Object Model

3000

PAPER ADD-->IAS OFFICER, 10+YEARS

```
title = document.querySelector('title')
```

```
links = document.querySelectorAll('a')
```

```
for (link of links){
```

```
    link.textContent='Sunny Leone';
```

```
    link.style.color='white';
```

```
    link.style.background='red'
```

```
}
```

Browser will constructor DOM. All HTML tags will be stored as javascript objects.

If javascript required any html component, then it will contact DOM, which provides the corresponding element.

To display document to the console:

document

To display dom objects on the console:

console.dir(document)

How to grab HTML elements from the DOM:

Important DOM attributes for this:

document.URL--->This is the original url of the website

document.body--->It returns everything inside body

document.head--->It returns head of the page

document.links----->It returns list of all links on the page

Important methods of DOM:

To grab html components, we have the following methods.

document.getElementById()

Returns element with the specified id

document.getElementsByClassName()

Returns list of elements belongs to the specified class

document.getElementsByTagName()

Returns list of all elements with the specified tag

document.querySelector()

Returns the first object matching CSS style selector

document.querySelectorAll()

Returns aa objects matching CSS style selector

<h1> ---1

<h2> --->2

** --->9**

class first--->2

id special--->1

```
.first{  
  color:red;  
}
```

```
#special{  
  color:blue;  
}
```

```
h1+p{  
}
```

```
document.getElementById('special')  
document.getElementsByClassName('first')  
document.getElementsByClassName('first')[0]
```

```
document.getElementsByClassName('first')[0]  
document.getElementsByTagName('h2')
```

```
document.querySelector('#special')  
document.querySelector('.first')  
document.querySelectorAll('.first')  
document.querySelectorAll('.first')[0]
```

```
document.querySelector('h1+p')  
document.getElementsByTagName('p')
```

if i hv both id as well as class for tag , then what to take in queryselector()?

```
<h1>This is h1 tag data</h1>
<p id='first' class='special'>This is first paragraph</p>
<p class='special'>This is second paragraph</p>
<p>This is third paragraph</p>
<p id='last'>This is fourth paragraph</p>
```

To grab first paragraph:

1. document.getElementById('first')
2. document.getElementsByClassName('special')[0]
3. document.getElementsByTagName('p')[0]
4. document.querySelector('#first')
5. document.querySelector('.special')
6. document.querySelectorAll('#first')[0]
7. document.querySelectorAll('.special')[0]
8. document.querySelector('h1+p')
9. document.querySelectorAll('h1+p')[0]
10. document.querySelector('p#first')

document.getElementById()

Returns element with the specified id

document.getElementsByClassName()

Returns list of elements belongs to the specified class

document.getElementsByTagName()

Returns list of all elements with the specified tag

document.querySelector()

Returns the first object matching CSS style selector

document.querySelectorAll()

Returns aa objects matching CSS style selector

Color Changer Application:

'0123456789ABCDEF'--->16

Math.random()--->from [0,1)

#035efc

0*16--->0

max--->0.99*16--->15.84

floor(15.84)--->15

test.html:


```
-----
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>This is My Title</title>
  </head>
  <body>
    <h1>Important Methods and Attributes of DOM</h1>
    <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do
    eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim
    ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut
    aliquip ex ea commodo consequat. Duis aute irure dolor in
    reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
    pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa
    qui officia deserunt mollit anim id est laborum.</p>
    <h2>Favourite Food:</h2>
    <ul>
      <li class='first'>Chicken</li>
      <li>Mutton</li>
      <li>Fish</li>
      <li id='special'>Any animal including human being</li>
    </ul>
    <h2>Favourite Drink:</h2>
    <ul>
      <li class='first'>KingFisher</li>
      <li>Knock out</li>
      <li>Milk</li>
```

```

    <li>Thumbsup</li>
    <li>Blood</li>
</ul>

<p>Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do
eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim
ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut
aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa
qui officia deserunt mollit anim id est laborum.</p>

<a href="https://www.youtube.com/durgasoftware">View
profile</a>

<script type='text/javascript' src='test.js'></script>
</body>
</html>

```

test.js:

```

var h1 = document.querySelector('h1')
var p1 = document.querySelectorAll('p')[0]
var p2 = document.querySelectorAll('p')[1]
var h21 = document.querySelectorAll('h2')[0]
var h22 = document.querySelectorAll('h2')[1]
var ul1 = document.getElementsByTagName('ul')[0]
var ul2 = document.getElementsByTagName('ul')[1]

```

```

function getRandomColor(){

```

```

letters = '0123456789ABCDEF'
color = '#'
for (var i = 0; i < 6; i++) {
    r = Math.floor(Math.random()*16)
    color = color+letters[r]
}
return color
}

function changeColor(){
    h1.style.color = getRandomColor()
    p1.style.color = getRandomColor()
    p2.style.color = getRandomColor()
    h21.style.color = getRandomColor()
    h22.style.color = getRandomColor()
    ul1.style.color = getRandomColor()
    ul2.style.color = getRandomColor()
    document.body.style.background=getRandomColor()
}
setInterval(changeColor,1000)

```

How to change color of element:

```
element.style.color='red'
```

DOM for content interaction:

BY using DOM, we can change text,html code and attributes.

How to change text:

```
var myh1 = document.querySelector('h1')  
myh1.textContent='DURGA SOFTWARE SOLUTIONS'
```

```
var myh2 = document.querySelector('h2')  
myh2.textContent='<a  
href="https://www.youtube.com/durgasoftware">Free Videos</a>'
```

How to change inner HTML Code:

```
var myh2 = document.querySelector('h2')  
myh2.innerHTML='<a  
href="https://www.youtube.com/durgasoftware">Free Videos</a>'
```

How to change attributes:

```
element.getAttribute('attributeName')  
anchor = document.querySelector('a')  
anchor.getAttribute('href')  
anchor.setAttribute('href','https://www.durgasoftonline.com')
```

```
document.body.style.background='red'
```

How to change logo:

`https://www.durgasoftonline.com/logo.png`

`logo = document.querySelector('img')`

`logo.setAttribute('src','https://www.durgasoftonline.com/logo.png')`

`link = document.querySelector('a')`

`link.setAttribute('href','https://www.durgasoftonline.com')`

Event Handling By using DOM:

Event--->something which is happend

eg: button click, dblclick,mouseover, mouseout

Listener--->listens events

whenever a particular event occurs automatically listener will perform required activity.

How to implement event handling:

`element.addEventListener(event,function)`

eg1: To change color of h1 on single click operation:

`myh1 = document.querySelector('h1')`

```
myh1.addEventListener('click',colorChanger)
```

eg2: To change color of h1 on Double click operation:

```
myh1 = document.querySelector('h1')  
myh1.addEventListener('dblclick',colorChanger)
```

eg3: To change color of h1 on mouse over operation:

```
myh1 = document.querySelector('h1')  
myh1.addEventListener('mouseover',colorChanger)
```

eg4: To change color of h1 on mouse out operation:

```
myh1 = document.querySelector('h1')  
myh1.addEventListener('mouseout',colorChanger)
```

eg5: To change content as HYDERABAD and color as blue on mouse over operation

To change content as BANGALORE and color as red on mouse out operation

```
myh1 = document.querySelector('h1')  
myh1.addEventListener('mouseover',function(){  
    myh1.textContent='HYDERABAD'
```

```
myh1.style.color='blue'  
})
```

```
myh1.addEventListener('mouseout',function(){  
  myh1.textContent='BANGALORE'  
  myh1.style.color='red'  
})
```

Body Color Changer Application:

test.html:

```
<!DOCTYPE html>  
<html lang="en" dir="ltr">  
  <head>  
    <meta charset="utf-8">  
    <title>This is My Title</title>  
  </head>  
  <body>  
    <h1>Body Color Changer Application</h1>  
    <button type="button" name="button">Click Here to change Body  
Color</button>  
    <script type='text/javascript' src='test.js'></script>  
  </body>  
</html>
```

test.js:

```
function getRandomColor(){  
  letters = '0123456789ABCDEF'  
  color = '#'  
  for (var i = 0; i < 6; i++) {  
    r = Math.floor(Math.random()*16)  
    color = color+letters[r]  
  }  
  return color  
}  
  
var b = document.querySelector('button')  
b.addEventListener('click',function(){  
  document.body.style.background=getRandomColor()  
})
```

Demo Application for Random Names and Random Colors:

mouseover and mouseout events

test.html:

```
<!DOCTYPE html>  
<html lang="en" dir="ltr">  
  <head>  
    <meta charset="utf-8">  
    <title>This is My Title</title>
```



```
</head>
<body>
  <h1>Name and Color Changer Application</h1>
  <script type='text/javascript' src='test.js'></script>
</body>
</html>
```

test.js:

```
function getRandomColor(){
  letters = '0123456789ABCDEF'
  color = '#'
  for (var i = 0; i < 6; i++) {
    r = Math.floor(Math.random()*16)
    color = color+letters[r]
  }
  return color
}

function getRandomName(){
  names = ['SUNNY
LEONE','MALLIKA','KATRINA','KAREENA','DEEPIKA','PRIYANKA','SAMA
NTHA']
  index = Math.floor(Math.random() * names.length)
  return names[index]
}

myh1 = document.querySelector('h1')
```

```
myh1.addEventListener('mouseover',function(){
    myh1.textContent = getRandomName()
    myh1.style.color = getRandomColor()
})
myh1.addEventListener('mouseout',function(){
    myh1.textContent = getRandomName()
    myh1.style.color = getRandomColor()
})
```

Q. Names should be displayed with specified color only like Katrina always with red color.

```
function getRandomColor(){
    letters = '0123456789ABCDEF'
    color = '#'
    for (var i = 0; i < 6; i++) {
        r = Math.floor(Math.random()*16)
        color = color+letters[r]
    }
    return color
}
function getRandomName(){
    names = ['SUNNY
LEONE','MALLIKA','KATRINA','KAREENA','DEEPIKA','PRIYANKA','SAMA
NTHA']
    index = Math.floor(Math.random() * names.length)
    return names[index]
}
```

```
myh1 = document.querySelector('h1')
myh1.addEventListener('mouseover',function(){
  name = getRandomName()
  if (name == 'SUNNY LEONE') {
    color='pink'
  } else if (name == 'MALLIKA') {
    color='orange'
  }
  else if (name == 'KATRINA') {
    color='red'
  }
  else if (name == 'KAREENA') {
    color='blue'
  }
  else if (name == 'DEEPIKA') {
    color='cyan'
  }
  else if (name == 'PRIYANKA') {
    color='magenta'
  }
  else if (name == 'SAMANTHA') {
    color='green'
  }
  myh1.textContent = name
  myh1.style.color = color
})
```

```
myh1.addEventListener('mouseout',function(){
  name = getRandomName()
  if (name == 'SUNNY LEONE') {
    color='pink'
  } else if (name == 'MALLIKA') {
    color='orange'
  }
  else if (name == 'KATRINA') {
    color='red'
  }
  else if (name == 'KAREENA') {
    color='blue'
  }
  else if (name == 'DEEPIKA') {
    color='cyan'
  }
  else if (name == 'PRIYANKA') {
    color='magenta'
  }
  else if (name == 'SAMANTHA') {
    color='green'
  }
  myh1.textContent = name
  myh1.style.color = color
})
```

<h1>Name and Color Changer Application</h1>

myh1.textContent='DURGASOFT'

<h1>DURGASOFT</h1>

myh1.innerHTML='<h2>Hello Friends</h2>'

<h2>Hello Friends</h2>

TIC TAC TOE Implementation:

On empty cell-->If we click then X should be placed.

If cell contains already X-->if we click then O should be placed.

If cell contains already O-->if we click then empty that cell.

If we click restart button, then all cells will be cleared.

ttt.html:

<!DOCTYPE html>

<html lang="en" dir="ltr">

<head>

<meta charset="utf-8">

<title>DURGASOFT TTT Game</title>

<link rel="stylesheet" href="tt.css">

```

<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@4.6.0/dist/css/bootst
rap.min.css" integrity="sha384-
B0vP5xmATw1+K9KRQjQERJvTumQW0nPEzvF6L/Z6nronJ3oUOFUFpCj
EUQouq2+I" crossorigin="anonymous">
</head>
<body>
<div class="container">
  <div class="jumbotron">
    <h1>DURGASOFT TIC TAC TOE GAME</h1>
    <p>Be Ready to play to Improve Logical thinking</p>
    <button id="b" type="button" name="button" class="btn btn-
primary btn-large">Restart Game</button>
  </div>
</div>
<table>
  <tr>
    <td></td>
    <td></td>
    <td></td>
  </tr>
  <tr>
    <td></td>
    <td></td>
    <td></td>
  </tr>
  <tr>
    <td></td>
  </tr>

```

```
<td></td>
```

```
<td></td>
```

```
</tr>
```

```
</table>
```

```
<script type="text/javascript" src='ttt.js'></script>
```

```
</body>
```

```
</html>
```

tt.css:

```
.container .jumbotron{
```

```
background: red;
```

```
color:yellow;
```

```
text-align: center;
```

```
}
```

```
table{
```

```
margin: auto;
```

```
}
```

```
td{
```

```
height: 100px;
```

```
width: 100px;
```

```
border: 5px solid red;
```

```
font-size: 50px;
```

```
text-align: center;
```

```
}
```

ttt.js:

```
var restartbutton = document.getElementById('b')
var cells = document.querySelectorAll('td')
function clearAllCells(){
  for (var i = 0; i < cells.length; i++) {
    cells[i].textContent=""
  }
}
restartbutton.addEventListener('click',clearAllCells)
```

```
function changeContent(){
  content=this.textContent
  if (content == '') {
    this.textContent='X'
  }
  else if (content=='X') {
    this.textContent='O'
  }
  else if (content == 'O') {
    this.textContent=""
  }
}
for (cell of cells) {
  cell.addEventListener('click',changeContent)
}
```


For mini project on ttt developement:

<https://www.geeksforgeeks.org/simple-tic-tac-toe-game-using-javascript/>

HTML

CSS

Bootstrap--->Ready made styles which can be used directly

JS

DOM

```
var h1tags = document.querySelectorAll('h1')
for(h1 in h1tags){
  h1.style.color='red'
}
```

jQuery:

jquery.com

jQuery is a fast, small, and feature-rich JavaScript library. It makes things like HTML document traversal and manipulation, event

handling, animation, and Ajax much simpler with an easy-to-use API that works across a multitude of browsers. With a combination of versatility and extensibility, jQuery has changed the way that millions of people write JavaScript.

The main advantage of jquery is, it provides several methods and objects in the form of javascript file, so that developer can use directly and his life will be simplified.

plain old javascript code: Vanilla javascript.

Sample code to change color of every h1 tag:

```
-----  
var allh1s = document.querySelectorAll('h1')  
for (var h1 of allh1s) {  
  h1.style.color='red'  
}
```

jQuery:

```
-----  
$('h1').css('color','red')
```

connect jQuery to html:

2 ways

1. locally or 2. By CDN

Limitations of jQuery:

1. jQuery won't do any extra activity.
2. performance problems

Because of these limitations, some part of developers community won't recommend jQuery usage.

youmightnotneedjquery.com

jQuery Selectors:

Vanilla Java script:

- getElementById()
- getElementsByClassName()
- getElementsByTagName()
- querySelector()
- querySelectorAll()

in jQuery we have only one way to select html element: \$ symbol

\$ ==>querySelectorAll()

\$('h1') ==>document.querySelector('h1')

Manipulate HTML Elements:

element selector

id selector

class selector

* selector

Descendant selector

Adjucent selector

attribute selector

nth-of-type selector

Once we grab elements by using \$ symbol, we can manipulate by using css() method.

`$(selector).css(property,value)`

`$('#h1').css('color','white')`

`$('#h1').css('background','red')`

`$('#h1').css('border','5px solid green')`

We can save selected html element by using variable.

`var x = $('#h1')`

`x.css('color','white')`

`x.css('background','red')`

```
x.css('border','5px solid green')
```

Instead of passing parameters one by one, we can create object and pass that object directly.

```
var x = $('h1')
var myCSS={
  color: 'white',
  background: 'green',
  border: '5px solid red'
}
x.css(myCSS)
```

```
$('h1').css({
  color: 'white',
  background: 'green',
  border: '5px solid red'
})
```

*****Note:** We can use `$()` to select html elements, and `css()` to manipulate html elements.

```
$('h1')--->querySelectorAll()
```

How to select a particular HTML element instead of all matched elements:

`$('element')`---->All matched html elements.

`$('element:first')`--->only first matched

`$('element:last')`--->only last matched

`$('element:first-of-type')`--->only first matched

`$('element:nth-of-type(2)')`--->only 2nd matched

`$('element:nth-of-type(even)')`--->only evenly matched

`$('element:nth-of-type(odd)')`--->only odd matched

eg:

`$('h1').css('color','red')`--->Select all h1 tags

`$('h1:first').css('color','red')`--->selects only first h1

`$('h1').first().css('color','red')`--->selects only first h1

`$('h1:first-of-type').css('color','red')`--->selects only first h1

`$('h1:nth-of-type(1)').css('color','red')`--->selects only first h1

`$('h1:nth-of-type(3)').css('color','red')`--->selects only 3rd h1

`$('h1:last').css('color','red')`--->selects only last h1

`$('h1').last().css('color','red')`--->selects only last h1

`$('h1:nth-of-type(even)').css('color','red')`--->selects only even h1

`$('h1:nth-of-type(odd)').css('color','red')`--->selects only odd h1

Q1. Write Vanilla java script and jQuery code to change all h1 tags text color as white and background as red?

Vanilla java script:

```
var allh1= document.querySelectorAll('h1')
for(h1 of allh1){
    h1.style.color='white';
    h1.style.background='red';
}
```

jQuery code:

```
var mystyle={
    color:'white',
    background:'red'
}
$('h1').css(mystyle)
```

Instead of this

```
$('h1').css({
    color:'white',
    background:'red'
})
```

Q2. Write vanilla javascript and jquery code to set all li tags font size as 20px?

Vanilla java script:

```
var allli= document.querySelectorAll('li')
```

```
for(li of allli){  
    li.style.fontSize='50px';  
}
```

jQuery:

```
$('.li').css('fontSize','50px')
```

Demo Application:

demo.html:

```
<!DOCTYPE html>  
<html lang="en" dir="ltr">  
  <head>  
    <meta charset="utf-8">  
    <title></title>  
    <script src="https://code.jquery.com/jquery-3.6.0.slim.min.js"  
integrity="sha256-  
u7e5khyithlldTpu22PHhENmPcRdFiHRjhAuHcs05RI="  
crossorigin="anonymous"></script>  
  </head>  
  <body>  
    <p> This is the first paragraph</p>  
    <p id='second'> This is the second paragraph</p>  
    <p class="remaining"> This is the third paragraph</p>
```



```
<p class="remaining"> This is the fourth paragraph</p>
</body>
</html>
```

Q1. Select all p tags and set background as green?

```
$('p').css('background','green')
```

Q2. Select all p tags with class 'remaining' and make them 200px width

```
$('p').css('background','green')
$('.remaining').css('width','200px')
```

Q3. Select all <p> tags with id='second' and give red solid 10px border?

```
$('#second').css('border','10px solid red')
```

Q4. Select only third <p> tag and change font size as 30px?

```
$('p:nth-of-type(3)').css('fontSize','30px')
```

The Most commonly used jQuery Methods:

1. text()--->textContent
2. html()
3. attr()
4. val()
5. addClass()
6. removeClass()

7. toggleClass()

1. text():

We can use this method to get or set text content of matched elements.

It acts as both getter and setter method.

text()---->getter method to get content

text(content)--->setter method to set content.

eg-1: To get content of h1 tag

```
$('h1').text()
```

```
"This is h1 tag"
```

eg-2: To get text of all li tags

To set h1 content as 'DURGASOFT'

```
$('h1').text('DURGASOFT')
```

2. html()

Get the HTML contents of the first element in the set of matched elements or set the HTML contents of every matched element.

`$('h1').html()`

`$('li').html()`

`$('li').html('<a href="https://amazon.com">Amazon</a>')`

`<a href='https://amazon.com'>Amazon</a>`

3. attr():

attr --->attribute

Get the value of an attribute for the first element in the set of matched elements or

set one or more attributes for every matched element.

attr(attributename) --->To get the value of the specified attribute of the first matched element.

attr(attributename,attributevalue)--->If the specified attribute already available then old value will be replaced with new value in every matched element.

If the specified attribute is not already available then this new attribute will be set.

`element.text()`--->returns text content of specified html

`element`(getter method)

`element.text('newcontent')`--->sets newcontent (setter method)

`element.html()`--->get inner html of specified element(getter method)

`element.html('html code')`--->set inner html of specified element(setter)

`element.attr(attributename)`--->Returns the value associated with sepcified attribute in the given html element.(getter method)

`element.attr(attributename,attributevalue)`--->setter method

`<img src='durga.jpg'>`

`$('#img').attr('src')`--->durga.jpg

`$('#img').attr('src','sunny.jpg')`

`$('#img').attr('height',50px)`

`$('#img').css({height:'150px',width:'150px'})`

1. To get src attribute value of the first image:

`$('img').attr('src')`

2. To set src attribute of all images with our new image.

`$('img').attr('src','https://images.unsplash.com/photo-`

`1583499871880-`

`de841d1ace2a?ixid=MnwxMjA3fDB8MHxzZWFyY2h8NDV8fGFuaW1h`

`bHN8ZW58MHx8MHx8&ixlib=rb-`

`1.2.1&auto=format&fit=crop&w=500&q=60')`

3. To change only first image src attribute value:

`$('img').first().attr('src','https://images.unsplash.com/photo-`

`1583499871880-`

`de841d1ace2a?ixid=MnwxMjA3fDB8MHxzZWFyY2h8NDV8fGFuaW1h`

`bHN8ZW58MHx8MHx8&ixlib=rb-`

`1.2.1&auto=format&fit=crop&w=500&q=60')`

Get the value of an attribute for the first element in the set of matched elements or set one or more attributes for every matched element

`$('img')`

`$('#first')`

In css:

1. element selectors
2. id selectors
3. class selectors
- 4> * selector
5. adjacent selector
6. descendant selector
7. nth of type selector

Demo for getting and setting type attribute of input tag:

User Input: <input type='text' name="" value="">

tag name: input

attribute name: type

eg-1: To get value of type attribute of input tag

`$('input').attr('type')`

eg-2: To set the value of type attribute with color:

`$('input').attr('type','color')`

eg-3: To set the value of type attribute with 'checkbox':

`$('input').attr('type','checkbox')`

- 1. text()**
- 2. html()**
- 3. attr()**
- 4. val():**

Get the current value of the first element in the set of matched elements or set the value of every matched element.

eg:

value entered in the text box

which radio button selected

which value selected from the dropdown box

Enter Name: <input type='text' name='user1' >

Enter Name: <input type='text' name='user1' >

- 1. To get value provided by enduser for the first input tag?**

`$('input').val()`

2. To get value provided by enduser for the second input tag?

`$('#input:nth-of-type(2)').val()`

3. To set for all input tags as sunny as the value

`$('#input').val('sunny')`

`$('#input:nth-of-type(2)').val('sunny')`

`$('#select').val('datascience')`

`addClass()`--->To add specified class to the set of matched elements

`removeClass()`--->To remove specified class from the set of matched elements

`toggleClass()`--->add/remove

css class

eg-1: Add high class for every li tag

`$('#li').addClass('high')`

eg-2: For every even li, remove high class

```
$('li:nth-of-type(even)').removeClass('high')
```

eg-3: For 2nd li add class completed

```
$('li:nth-of-type(2)').addClass('completed')
```

```
$('li:nth-of-type(2)').addClass('low')
```

eg-4: toggle class high for every li tag:

```
$('li').toggleClass('high')
```

Event Handling by using jQuery:

To make html elements interactive.

jQuery defines several methods for event handling.

<https://api.jquery.com/category/events/>

3 top most commonly used methods in jQuery Event handling:

1. click()

2. keypress()

3. on()--->general purpose method

1. click():

We can use click() method to add click listener to the element.

eg-1: To raise alert message when ever we click h1 tag?

```
$('#h1').click(function(){  
  alert('h1 tag got clicked')  
})
```

eg-2: To raise alert message and to change background color of button on click event

```
$('#button').click(function(){  
  alert('Hello dont sleep');  
  $(this).css('background','red');  
})
```

demo.html:

```
<!DOCTYPE html>  
<html lang="en" dir="ltr">  
  <head>  
    <meta charset="utf-8">  
    <title></title>  
    <script src="https://code.jquery.com/jquery-3.6.0.slim.min.js"  
integrity="sha256-
```

```

u7e5khyithlIdTpu22PHhENmPcRdFiHRjhAuHcs05RI="
crossorigin="anonymous"></script>
</head>
<body>
  <h1>jQuery Event Handling Demo</h1>
  <button type="button" name="button">Dont Sleep First
Warning</button>
  <button type="button" name="button">Dont Sleep Second
Warning</button>
  <button type="button" name="button">Dont Sleep Third
Warning</button>
  <script type="text/javascript" src='demo.js'></script>
</body>
</html>

```

demo.js:

```

-----
$('h1').click(function(){
  alert('Click any button, you will get greetings message')
})

$('button:first').click(function(){
  alert('Hello Dont Sleep')
  $(this).css('background','yellow')
})

$('button:nth-of-type(2)').click(function(){

```

```
    alert('Hello Dont Sleep, I will beat you')
    $(this).css('background','orange')
  })
```

```
$('#button:last').click(function(){
    alert('Hello Stupid, dont sleep, I will kill you')
    $(this).css('background','red')
  })
```

2. keypress():

We can use this method to add key press listener to elements.

eg:

Enter Name: <input type='text'>

character --->key code

x or X

X 88

x IS 120

```
$('#input').keypress(function(event){
```

```
    console.log(event)
  })
```

```
$('#input').keypress(function(event){
  if(event.which == 88 || event.which == 120){
    alert('Hello you are pressing x or X, You are under monitoring!!!')
  }
})
```

To raise alert message when ever we press enter key:

```
-----
$('#input').keypress(function(event){
  if(event.which == 13 ){
    alert('Hello you just pressed enter key and your typed content
is:'+$ (this).val())
  }
})
```

3. on():

It is similar to vanilla javascript addEventListener()

eg-1: Whenever mouseover event happend, the text of h1 tag should be changed to 'Bangalore City' with red background and white text?

```
$('#h1').on('mouseover',function(){  
    $(this).text('BANGALORE CITY')  
    $(this).css({background:'red',color:'white',fontSize:'25px'})  
})
```

```
$('#h1').on('mouseout',function(){  
    $(this).text('HYDERABAD CITY')  
    $(this).css({background:'green',color:'white',fontSize:'50px'})  
})
```

eg: for button:

single click: alert message as 'Hello Stupid dont klik me!!!'

Double Click: alert message as 'Hello Animal, I will kill you!!!'

```
$('#button:first').on('click',function(){  
    alert('Hello Stupid, dont klik me!!!')  
})
```

```
$('#button:last').on('dblclick',function(){  
    alert('Hello Animal, I will kill you!!!')  
})
```

jQuery Effects:

jQuery provides several inbuilt effects. The main important effects are:

1. Fading effects
2. Sliding effects

1. Fading effects:

fadeOut():

hide the matched elements by fading them to transparent.

fadeIn():

Display the matched elements which are fadeout

fadeToggle()

Fadeout--->FadeIn

FadeIn--->FadeOut

```
$('button').on('click',function(){  
    $('div').fadeOut(2000);  
})
```

```
$('button').on('click',function(){  
    $('div').fadeToggle(3000);  
})
```

2. Sliding effects:

.slideDown()

Display the matched elements with a sliding motion.

.slideToggle()

Display or hide the matched elements with a sliding motion.

.slideUp()

Hide the matched elements with a sliding motion.

Bootstrap:

xs removed.

jquery: until 3.5 version